Supplemental Material for "Enabling quantum communication in ultra-large-scale networks"

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QUANTUM COMMUNICATION NETWORKS AND OPERATIONS OF ENTANGLEMENT DISTRIBUTION

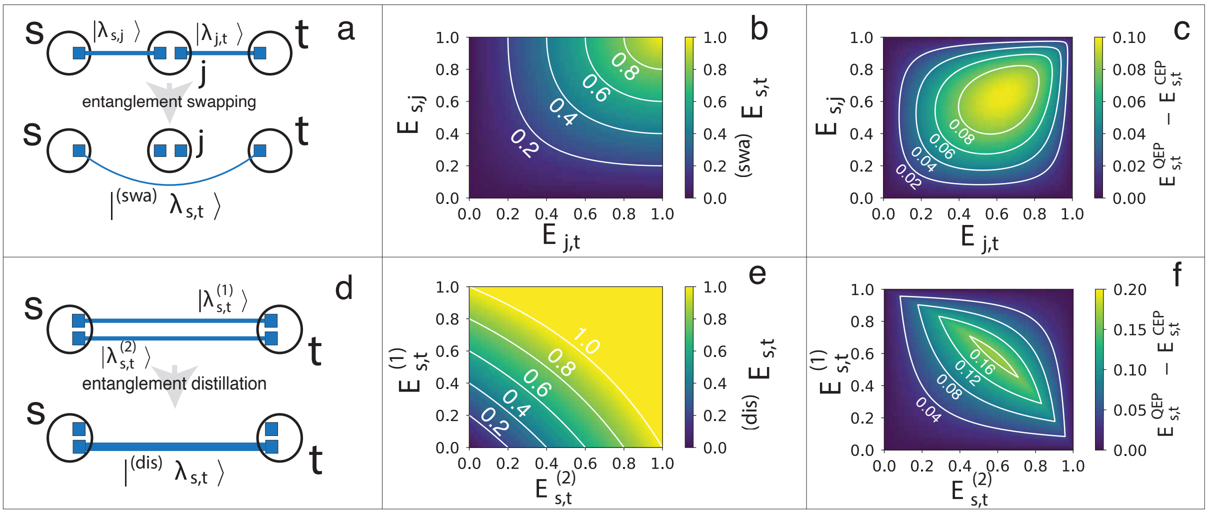

Figure S1. **Operations of entanglement distribution.** (a) In this schematic representation, black circles represent nodes in the quantum network and blue squares represent qubits located on the nodes. Blue segments denote entangled qubits, and their thickness is proportional to the qubits' entanglement. In the operation of entanglement swapping, two states $|\lambda_{s,j}\rangle$ and $|\lambda_{j,t}\rangle$ in series are mapped to new a state $|^{(\mathrm{swa})}\lambda_{s,t}\rangle$. (b) Entanglement $^{(\mathrm{swa})}E_{s,t} = E(|^{(\mathrm{swa})}\lambda_{s,t}\rangle)$ as a function of the entanglement of the states $|\lambda_{s,j}\rangle$ and $|\lambda_{j,t}\rangle$ resulting from the operation of entanglement swapping. To make the heatmap more readable, we display also some contour curves. (c) We compare the entanglement between nodes s and t after, i.e., $E_{s,t}^{\mathrm{QEP}} = E(|^{(\mathrm{swa})}\lambda_{s,t}\rangle)$, and before, i.e., $E_{s,t}^{\mathrm{CEP}} = E(|\lambda_{s,j}\rangle)E(|\lambda_{j,t}\rangle) = E_{s,j}E_{j,t}$, performing the operation of entanglement swapping. Such an entanglement differential is plotted as a function of the entanglement of the states $|\lambda_{s,j}\rangle$ and $|\lambda_{j,t}\rangle$. (d) Thanks to the operation of entanglement distillation, two parallel states $|\lambda_{s,t}^{(1)}\rangle$ and $|\lambda_{s,t}^{(2)}\rangle$ create a new state $|^{(\mathrm{dis})}\lambda_{s,t}\rangle$. (e) Entanglement $^{(\mathrm{dis})}E_{s,t} = E(|^{(\mathrm{dis})}\lambda_{s,t}\rangle)$ as a function of the entanglement of the states $|\lambda_{s,t}^{(1)}\rangle$ and $|\lambda_{s,t}^{(2)}\rangle$ resulting from the operation of entanglement distillation. (f) We compare the entanglement between the two nodes s and t in panel d after, i.e., $E_{s,t}^{\mathrm{QEP}} = E(|^{(\mathrm{dis})}\lambda_{s,t}\rangle)$, and before, i.e., $E_{s,t}^{\mathrm{CEP}} = 1-\left[1 - E(|\lambda_{s,t}^{(1)}\rangle)\right]\left[1 - E(|\lambda_{s,t}^{(2)}\rangle)\right] = 1-\left(1 - E_{s,t}^{(1)}\right)\left(1 - E_{s,t}^{(2)}\right)$, performing the operation of entanglement distillation. Such an entanglement differential is plotted as a function of the entanglement of the states $|\lambda_{s,t}^{(1)}\rangle$ and $|\lambda_{s,t}^{(2)}\rangle$.

A quantum communication network can be seen as a classical network, where pairs of nodes are connected via weighted (multi-) edges [1, 2]. A node stands for a station containing an arbitrary number of qubits; an edge between two nodes represents a quantum channel, i.e., an entangled state between a pair of qubits; the weight of an edge reflects the entanglement between qubits; finally, multi-edges can connect the very same pair of nodes if multiple channels between the two nodes exist, see Fig. S1. This representation is very useful for the analysis of protocols enabling communication between far apart non-entangled stations as the elementary quantum operations at the basis of the propagation of information can be mapped to edge transformations on the classical graph [1], see Fig. S1.

To be more specific, the edge (i, j) between the nodes i and j stands for a pair of entangled qubits that in the Schmidt form is written as

$$|\lambda_{i,j}\rangle = \sqrt{\lambda_{i,j}}\,|00\rangle + \sqrt{1-\lambda_{i,j}}\,|11\rangle \tag{S1}$$

where the Schmidt coefficient $\lambda_{i,j} \in [1/2, 1]$. In the notation $|xy\rangle$, $x = 0, 1$ and $y = 0, 1$ respectively denote the possible states of the qubits located on the nodes at the end of the edge [3]. Entanglement between the two qubits is apparent from the presence of the only terms $|00\rangle$ and $|11\rangle$, and the absence of the terms $|01\rangle$ and $|10\rangle$. This means that, by taking a measurement in the $|0\rangle$'s basis or in the $|1\rangle$'s basis from either node, the state $|\lambda_{i,j}\rangle$ collapses randomly in either $|00\rangle$ or $|11\rangle$. As a metric of entanglement, we use

$$E(|\lambda_{i,j}\rangle) = 2(1 - \lambda_{i,j}) \, , \tag{S2}$$

i.e., the probability to convert $|\lambda_{i,j}\rangle$ to a singlet or Bell state [1]. $\lambda_{i,j} = \frac{1}{2}$ indicates maximal entanglement, whereas $\lambda_{i,j} = 1$ denotes non-entangled qubits. We stress that different notations are used in the literature. The most common parameterizations are in terms of the angle $\theta_{i,j} \in [0, \pi/4]$, the concurrence $c_{i,j} \in [0, 1]$ or the Schmidt coefficient $\lambda_{i,j} \in [\frac{1}{2}, 1]$. All these notations are equivalent upon proper transformations, i.e., $c_{i,j} = \sin 2\theta_{i,j} = 2\sqrt{\lambda_{i,j}}\sqrt{1-\lambda_{i,j}}$.

In their seminal paper [1], Acin *et al.* showed that the above model has an immediate mapping into the classical problem of network percolation where the occupation probability of each edge is given by Eq. (S2), and the probability that two nodes can communicate is given by the existence of at least one path of occupied edges connecting the two nodes [4, 5]. According to this mapping named Classical Entanglement Percolation (CEP), the existence of the giant connected component in the classical network represents a sufficient condition for system-wide communication in the quantum network. In particular, if all edges are characterized by the same entanglement, then the percolation threshold of the classical network maps to the minimal entanglement making CEP a feasible protocol for communication.

At the same time, Acin *et al.* showed that the sufficient conditions for quantum communication predicted using CEP often represent loose bounds for the actual minimal requirements for network-wide quantum communication. Greatly improved protocols can be obtained by taking advantage of two local operations performed on the edges of the network, namely entanglement swapping and entanglement distillation. Both these operations are possible because of the quantum nature of the network. To contrast it to CEP, they referred to this improved approach as Quantum Entanglement Percolation (QEP).

Entanglement swapping is the operation that acts on two pairs of qubits, namely $|\lambda_{s,j}\rangle$ and $|\lambda_{j,t}\rangle$, placed in series in the network. The two states can be mapped into one pure state by performing a Bell measurement on the middle qubits [6, 7]. The entanglement of the new state depends on the details of the measurement, however, this is often simplified by only considering the entanglement of the worst-case scenario [8], according to which the newly generated state expressed in the Schmidt form of Eq. (S1) has associated coefficient

$$^{(\mathrm{swa})}\lambda_{s,t} = \frac{1+\sqrt{1-16\lambda_{s,j}(1-\lambda_{s,j})\lambda_{j,t}(1-\lambda_{j,t})}}{2} \,. \tag{S3}$$

Since the coefficient of Eq. (S3) corresponds to the minimally entangled outcome that is obtained from the operation of entanglement swapping, it can be safely considered as a deterministic transformation in protocols of communication on quantum networks [9–12]. The operation of entanglement swapping allows to create entanglement between states that are not initially entangled (Figure S1b); this happens at cost of decreasing entanglement as $E(|^{(\mathrm{swa})}\lambda_{s,t}\rangle) \leq \min\left\{E(|\lambda_{s,j}\rangle), E(|\lambda_{j,t}\rangle)\right\}$. On the other hand, the probability of creating a singlet between two nodes is enhanced by the operation of entanglement swapping compared with what predicted by CEP, i.e., $E^{\mathrm{QEP}}_{s,t} = E(|^{(\mathrm{swa})}\lambda_{s,t}\rangle) \geq E(|\lambda_{s,j}\rangle)\, E(|\lambda_{j,t}\rangle) = E^{\mathrm{CEP}}_{s,t}$, see Figure S1c. All these results generalize to edges along the path $\mathbf{p}_{s,t} = (s = j_0, j_1, \ldots, j_\ell = t)$, where entanglement swapping can be used to create a new state whose Schmidt coefficient equals

$$^{(\mathrm{swa})}\lambda_{s,t} = \frac{1+\sqrt{1-2^{2\ell}\prod_{j=1}^{\ell}\lambda_{j-1,j}(1-\lambda_{j-1,j})}}{2} \,. \tag{S4}$$

Still, we have that $E(|^{(\mathrm{swa})}\lambda_{s,t}\rangle) \leq \min_{j=1}^{\ell} E(|\lambda_{j-1,j}\rangle)$ and $E^{\mathrm{QEP}}_{s,t} = E(^{(\mathrm{swa})}\lambda_{s,t}) \geq \prod_{j=1}^{\ell} E(|\lambda_{j-1,j}\rangle) = E^{\mathrm{CEP}}_{s,t}$.

Entanglement distillation operates on two pairs of qubits disposed in parallel (i.e., a multi-edge in the jargon of network science) to create a single entangled state [13], see Figure S1d. In particular as proven in Refs. [14–16], the most entangled state that can be obtained deterministically by distilling two pairs of parallel qubits $|\lambda^{(1)}_{s,t}\rangle$ and $|\lambda^{(2)}_{s,t}\rangle$ is characterized by the Schmidt coefficient

$$^{(\mathrm{dis})}\lambda_{s,t} = \max\left\{\frac{1}{2}, \lambda^{(1)}_{s,t}\lambda^{(2)}_{s,t}\right\} \,. \tag{S5}$$

We see that $E(|^{(\mathrm{dis})}\lambda_{s,t}\rangle) \geq \max\{E(|\lambda^{(1)}_{s,t}\rangle), E(|\lambda^{(2)}_{s,t}\rangle)\}$, meaning that the operation of entanglement distillation improves the entanglement of each distilled state, see Figure S1e. One can even generate a maximally entangled state from two non-maximally entangled states. Entanglement distillation also shows an increase in the probability of creating a singlet compared to what predicted by CEP, i.e., $E^{\mathrm{QEP}}_{s,t} = E(|^{(\mathrm{dis})}\lambda_{s,t}\rangle) \geq 1-\left[1-E(|\lambda^{(1)}_{s,j}\rangle)\right]\left[1-E(|\lambda^{(2)}_{s,t}\rangle)\right] = E^{\mathrm{CEP}}_{s,t}$, see Figure S1f. The operation of entanglement distillation generalizes to k parallel edges $|\lambda^{(1)}_{s,t}\rangle$, $|\lambda^{(2)}_{s,t}\rangle$, …, $|\lambda^{(k)}_{s,t}\rangle$ by creating a new state with Schmidt coefficient equal to

$$^{(\mathrm{dis})}\lambda_{s,t} = \max\left\{\frac{1}{2}, \prod_{j=1}^{k}\lambda^{(j)}_{s,t}\right\} \,. \tag{S6}$$

Also here, $E(|^{(\mathrm{dis})}\lambda_{s,t}\rangle) \geq \max_{j=1}^{\ell} E(|\lambda^{(j)}_{s,t}\rangle)$ and $E^{\mathrm{QEP}}_{s,t} = E(^{(\mathrm{dis})}\lambda_{s,t}) \geq 1-\prod_{j=1}^{\ell}\left[1-E(|\lambda^{(j)}_{s,t}\rangle)\right] = E^{\mathrm{CEP}}_{s,t}$.

Limiting performance of the QEP protocol

Here, we provide a more explicit derivation of the results already illustrated in the main paper. Our focus is on estimating the performance of the QEP protocol between two nodes that are connected by k independent paths each of length ℓ. The minimum value of the Schmidt coefficient needed to establish a maximum entanglement is given by

$$\lambda^*(k,\ell) = \frac{1+\sqrt{1-2^{\frac{2k-1}{k\ell}}\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{\ell}}}}{2} \ . \tag{S7}$$

This result is achieved by first finding the coefficient of the state obtained by swapping ℓ states with coefficient λ, thus

$$^{(\mathrm{swa})}\lambda = \frac{1+\sqrt{1-2^{2\ell}\lambda^{\ell}(1-\lambda)^{\ell}}}{2}$$

and then performing the distillation of k of such states, i.e.,

$$^{(\mathrm{dis})}\lambda = \left[\frac{1+\sqrt{1-2^{2\ell}\lambda^{\ell}(1-\lambda)^{\ell}}}{2}\right]^k$$

and finally equating the previous expression to $\frac{1}{2}$, thus

$$\left\{\frac{1+\sqrt{1-[2^2\lambda^*(1-\lambda^*)]^{\ell}}}{2}\right\}^k = \frac{1}{2} \ . \tag{S8}$$

Then, one can perform some simple math manipulations

$$\frac{1+\sqrt{1-[2^2\lambda^*(1-\lambda^*)]^{\ell}}}{2} = \frac{1}{2^{1/k}}$$

$$1+\sqrt{1-[2^2\lambda^*(1-\lambda^*)]^{\ell}} = 2^{(k-1)/k}$$

$$\sqrt{1-[2^2\lambda^*(1-\lambda^*)]^{\ell}} = 1-2^{(k-1)/k}$$

$$1-\left[2^2\lambda^*(1-\lambda^*)\right]^{\ell} = (1-2^{(k-1)/k})^2 = 1-2^{(2k-1)/k}+2^{(2k-2)/k} = 1-2^{(2k-1)/k}\left(1-2^{-1/k}\right)$$

$$\left[2^2\lambda^*(1-\lambda^*)\right]^{\ell} = {}^{(2k-1)/k}\left(1-2^{-1/k}\right)$$

and

$$2^2\lambda^*(1-\lambda^*) = {}^{(2k-1)/(\ell k)}\left(1-2^{-1/k}\right)^{1/\ell} \ .$$

The latter expression is the square of the concurrence of the resulting channel, i.e.,

$$c^*(k,\ell) = 2^{\frac{2k-1}{k\ell}}\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{\ell}} \ , \tag{S9}$$

from which we get Eq. (S7). For fixed ℓ, $\lambda^*(k,\ell)$ grows as k increases, indicating that less entanglement at the individual edge level is required to establish a perfect quantum communication channel. In particular, we have that $\lim_{k\to\infty}\lambda^*(k,\ell) = 1$ if ℓ is finite. For fixed k, however, $\lambda^*(k,\ell)$ decreases as ℓ increases, denoting that entanglement degrades as paths become longer and longer. In particular, we have that $\lim_{\ell\to\infty}\lambda^*(k,\ell) = 1/2$ if k is finite. The actual interesting behavior occurs when both k and ℓ increase at the same time, as the behavior of $\lambda^*(k,\ell)$ is determined by the relative increase of ℓ and k.

To this end, let's consider $c^*(k,\ell)$. For the first term, we have $\lim_{k,\ell\to\infty} 2^{\frac{2k-1}{k\ell}} = 1$. To study the limit of the second term, we consider three main cases that are relevant for us. The first scenario is $k \sim \ell^{\alpha}$, where we can write

$$\lim_{k\to\infty}\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{k^{1/\alpha}}} = \exp\left\{\lim_{k\to\infty}\log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{k^{1/\alpha}}}\right]\right\}$$

then

$$\lim_{k\to\infty} \log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{k^{1/\alpha}}}\right] = \lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{k^{1/\alpha}} \ .$$

We notice that

$$\lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{k} = \lim_{k\to\infty} \frac{\log(\log 2) - \log k}{k^{1/\alpha}} = 0 \ ,$$

where we used the Taylor's series expansion $2^{-\frac{1}{k}} = e^{-\frac{1}{k}\log 2} \simeq 1 - \frac{1}{k}\log 2 + \cdots$ In summary, $\lim_{k\to\infty}\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{k^{1/\alpha}}} = 1$, hence,

$$\lim_{N\to\infty} \lambda^*(k,\ell) = \frac{1}{2} \tag{S10}$$

if k diverge as a power of ℓ as N increases. The same conclusion is valid also for k diverging slower than a power of ℓ to infinity as well as for finite k.

The next case is assuming k diverging at least exponentially with ℓ. There are two relevant cases for our purposes: (i) $k \sim N^\alpha$ and $\ell \sim \log N$, and (ii) $k \sim N^\alpha$ and $\ell \sim \log\log N$. Here, α is a free parameter determining how quickly k is diverging with N. For our networks, we should have $0 < \alpha \leq \min\{\frac{1}{2}, \frac{1}{\gamma-1}\}$. The different scaling of ℓ are instead useful for the case of small-world or ultra-small-world networks, respectively. Note that the additional case $k \sim (\log N)^\alpha$ and $\ell \sim \log\log N$ is basically the same as of case (i) upon re-interpretation of the parameter α. In case (i), we can write $\ell \sim \frac{1}{\alpha}\log k$, thus

$$\lim_{k\to\infty}\left(1-2^{-\frac{1}{k}}\right)^{\frac{\alpha}{\log k}} = \exp\left\{\lim_{k\to\infty}\log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{\alpha}{\log k}}\right]\right\}$$

then

$$\lim_{k\to\infty}\log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{\alpha}{\log k}}\right] = \alpha \lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{\log k} \ .$$

Once more we can use the the Taylor's series expansion $2^{-\frac{1}{k}} = e^{-\frac{1}{k}\log 2} \simeq 1 - \frac{1}{k}\log 2 + \cdots$ to get

$$\lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{\log k} = \lim_{k\to\infty} \frac{\log(\log 2) - \log k}{\log k} = -1 \ .$$

Putting things together, we finally can write that

$$\lim_{N\to\infty} \lambda^*(k,\ell) = \frac{1+\sqrt{1-e^{-\alpha}}}{2} \tag{S11}$$

if $k \sim N^\alpha$ and $\ell \sim \log N$ or if $k \sim (\log N)^\alpha$ and $\ell \sim \log\log N$.

For case (ii), we can write $\ell \sim \log\frac{1}{\alpha} + \log\log k$. Then, we can repeat similar steps as above:

$$\lim_{k\to\infty}\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{\log\frac{1}{\alpha}+\log\log k}} = \exp\left\{\lim_{k\to\infty}\log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{\log\log k}}\right]\right\}$$

then

$$\lim_{k\to\infty}\log\left[\left(1-2^{-\frac{1}{k}}\right)^{\frac{1}{\log\log k}}\right] = \lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{\log\log k} \ .$$

Once more we can use the the Taylor's series expansion $2^{-\frac{1}{k}} = e^{-\frac{1}{k}\log 2} \simeq 1 - \frac{1}{k}\log 2 + \cdots$ to get

$$\lim_{k\to\infty} \frac{\log\left(1-2^{-\frac{1}{k}}\right)}{\log k} = \lim_{k\to\infty} \frac{\log(\log 2) - \log k}{\log\log k} = -\infty \ .$$

In summary, we have that

$$\lim_{N\to\infty} \lambda^*(k,\ell) = 1 \tag{S12}$$

if $k \sim N^\alpha$ and $\ell \sim \log\log N$. Putting everything together, we have that

$$\lim_{\ell\to\infty} \lambda^*(k,\ell) = \begin{cases} 1 & \text{if } k \sim \exp[\exp(\alpha\ell)]\,,\ \forall\alpha > 0 \\ \frac{1+\sqrt{1-\exp(-\alpha)}}{2} & \text{if } k \sim \exp(\alpha\ell)\,,\ \forall\alpha \geq 0 \end{cases} \ . \tag{S13}$$

Using the relation $E^*_{\text{edge}}(k,\ell) = 2[1-\ell^*(k,\ell)]$, we find the limiting behavior for the minimum level of entanglement required for the individual edges of the network.

Relaxing the condition for maximal entanglement

We note that Eq. (S8) is obtained with the goal of determine the minimum level of entanglement at individual edges required to create a maximally entangled channel. We can relax this condition by writing

$$\left\{ \frac{1 + \sqrt{1 - [2^2\lambda^*(1-\lambda^*)]^\ell}}{2} \right\}^k = a \ . \tag{S14}$$

with $a \in [1/2, 1]$. Following similar calculations as above, we arrive to the analogous of Eq. (S9)

$$c^*(k,\ell) = \left(2^2 a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} \left(1 - a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} \ . \tag{S15}$$

For $a = 1/2$, we are exactly in the same conditions as in the section above. For $a = 1$, we trivially have $c^*(k,\ell) = 0$, i.e., no initial entanglement is required to reach a non-entangled quantum channel.

For $a \in (1/2, 1)$, we should study the limiting behavior of the two terms appearing on the right hand side of Eq. (S15). For the first term, we can write

$$\left(2^2 a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} = \exp\left\{ \log\left[\left(2^2 a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} \right] \right\} = \exp\left\{ \frac{1}{\ell} \left[2\log 2 + \frac{1}{k}\log a \right] \right\}$$

thus

$$\lim_{N\to\infty} \left(2^2 a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} = 1 \ ,$$

as long as $\ell \to_{N\to\infty} \infty$. For the second term, we can use simila calculations as in the section above to recover the very same results, i.e.,

$$\lim_{\ell\to\infty} \left(1 - a^{\frac{1}{k}}\right)^{\frac{1}{\ell}} = \begin{cases} 0 & \text{if } k \sim \exp[\exp(\alpha\ell)] \ , \ \forall \alpha > 0 \\ e^{-\alpha} & \text{if } k \sim \exp(\alpha\ell) \ , \ \forall \alpha \geq 0 \end{cases} \ . \tag{S16}$$

In summary, our results are for any desired level of entanglement for the channel greater than zero.

Configuration model

We consider networks generated according to the uncorrelated configuration model [17, 18]. To create an instance of the model with N nodes, we first extract N random variables from the degree distribution

$$\mathcal{K}^{(0)}(k) = \begin{cases} k^{-\gamma} & \text{if } k \in [k_m, k_M = \min\{N^{1/2}, N^{1/(\gamma-1)}\}] \\ 0 & \text{otherwise} \end{cases} \ . \tag{S17}$$

We then create edges between nodes at random but preserving the pre-imposed degree sequence. In the creation of the graphs, we suppress multi-edge and self-loops. This causes minor differences between the resulting degree sequence of the graph and the pre-imposed one, visible in an extremely small fraction of nodes with degree smaller than k_m, see for example Fig. S2.

All our numerical results are obtained by taking averages over 10 different network instances for each combination of the parameters N and γ. The only exception is the case $N = 10^8$ where we consider only one graph instance. In all cases, we set $k_m = 3$.

Analysis of quantum communication protocols in the configuration model

Denote with $\mathcal{K}^{(0)}(\leq k) = \sum_{x=k_m}^{k} \mathcal{K}^{(0)}(x)$ the cumulative of the degree distribution. Clearly, $\mathcal{K}^{(0)}(k) = \mathcal{K}^{(0)}(\leq k) - \mathcal{K}^{(0)}(\leq k-1)$.

Consider now a pair of randomly selected nodes s and t and define $k = \min\{k_s, k_t\}$ as the minimum of the degree of the two nodes. The cumulative distribution of k is

$$\mathcal{K}^{(0)}_{\min}(\leq k) = 1 - \left[1 - \mathcal{K}^{(0)}(\leq k)\right]^2 = 2\mathcal{K}^{(0)}(\leq k) - \left[\mathcal{K}^{(0)}(\leq k)\right]^2 \ . \tag{S18}$$

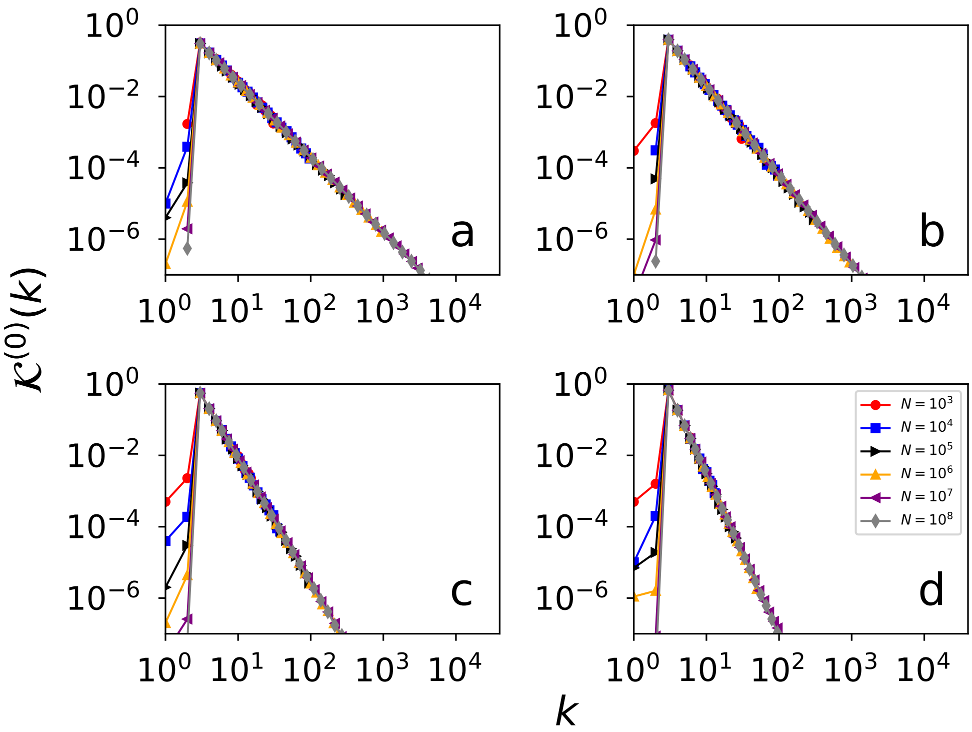

Figure S2. **Statistical properties of the configuration model.** (a) We plot the degree distribution $\mathcal{K}^{(0)}(k)$ for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

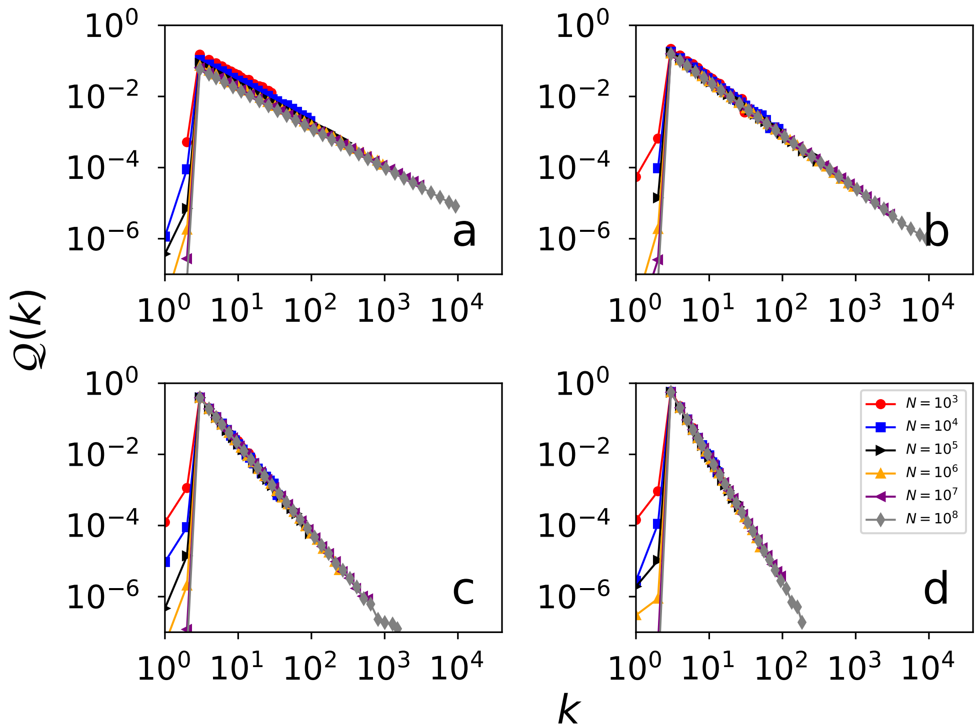

Figure S3. **Statistical properties of the configuration model.** (a) We plot the distribution of the degree of a neighbor of a randomly chosen node $\mathcal{Q}(k)$ for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

In fact, $\left[1-\mathcal{K}^{(0)}(\leq k)\right]^2$ is the probability that two random numbers extracted from $\mathcal{K}^{(0)}(k)$ are larger than k. This probability is just the complement of $\mathcal{K}^{(0)}_{\min}(\leq k)$. Pretend now that we select a node with degree k. If we look at the end of one its edges, we find a node with degree q which is a random variate obeying the cumulative distribution

$$Q(\leq q) = \frac{1}{\langle k\rangle}\sum_{x=k_m}^{q} x\,\mathcal{K}^{(0)}(x) = \frac{\sum_{x=k_m}^{q} x\,\mathcal{K}^{(0)}(x)}{\sum_{x=k_m}^{k_M} x\,\mathcal{K}^{(0)}(x)} \; . \tag{S19}$$

Numerical estimates of the distribution $Q(q)$ are displayed in Fig. S3. If we consider all the k neighbors, and get the maximum value of the observed degree, i.e., the variable $q = \max\{q_1, q_2, \ldots, q_k\}$, this variable will obey the cumulative distribution

$$\mathcal{K}^{(1)}(\leq q|k) = [Q(\leq q)]^k \; . \tag{S20}$$

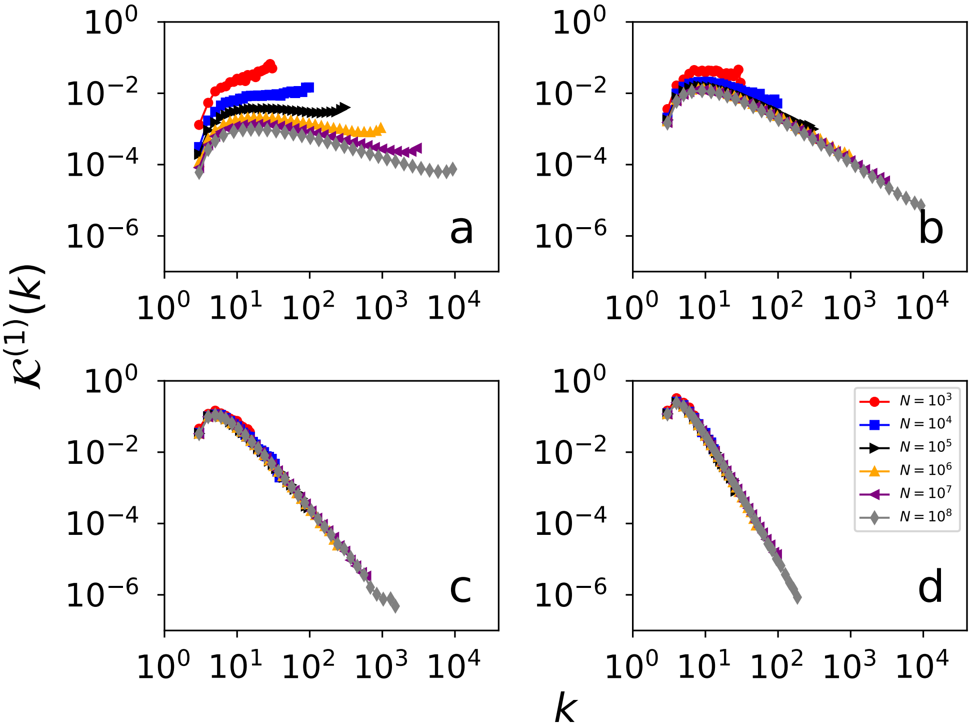

Figure S4. **Statistical properties of the configuration model.** (a) We plot the distribution of the maximum degree of all nodes at most distance $x = 1$ from a randomly chosen node. This is roughly the same as $\mathcal{K}^{(1)}(k)$. Different colors and symbols are for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

Specifically, the cumulative probability of selecting a random node and finding that the degree of its largest-degree neighbor is k is

$$\mathcal{K}^{(1)}(\leq k) = \sum_x \mathcal{K}^{(1)}(x)\,\mathcal{K}^{(1)}(\leq k|x) = G_0\left[Q(\leq k)\right] \; , \tag{S21}$$

i.e., the generating function $G_0(\cdot)$ of the degree distribution computed at $Q(\leq y)$. Numerical estimates of the distribution $\mathcal{K}^{(1)}(k)$ are displayed in Fig. S4. Suppose now that we have two nodes s and t with degrees k_s and k_t. We then find their largest-degree neighbors r_s and r_t. The degrees k_{r_s} and k_{r_t} of these nodes obey Eq. (S19). The cumulative of the variable $k = \min\{k_{r_s}, k_{r_t}\}$ is

$$\begin{aligned}\mathcal{K}^{(1)}_{\min}(\leq k|k_s, k_t) = \;& 1 - \left[1-\mathcal{K}^{(1)}(\leq k|k_s)\right]\left[1-\mathcal{K}^{(1)}(\leq k|k_t)\right] \\ & [Q(\leq k)]^{k_s} + [Q(\leq k)]^{k_t} - [Q(\leq k)]^{k_s+k_t}\end{aligned} \; . \tag{S22}$$

Hence, the cumulative probability that the minimum value of the degree of the largest-degree neighbors of two randomly extracted nodes is

$$\begin{aligned}\mathcal{K}^{(1)}_{\min}(\leq k) = \;& \sum_x \sum_y \mathcal{K}^{(1)}_{\min}(\leq k|x,y)\,\mathcal{K}^{(0)}(x)\,\mathcal{K}^{(0)}(y) \\ & 2\sum_x [Q(\leq k)]^x\,\mathcal{K}^{(0)}(x) - \left\{\sum_x [Q(\leq k)]^x\,\mathcal{K}^{(0)}(x)\right\}^2 \\ & 2G_0\left[Q(\leq k)\right] - \left\{G_0\left[Q(\leq k)\right]\right\}^2\end{aligned} \; . \tag{S23}$$

Finally, suppose we first select a random node whose degree is k, then we look at all next-nearest neighbors of this node. This is still a stochastic variable v that obeys the distribution

$$\mathcal{V}(v|k) = \sum_{q_1+q_2+\ldots q_k=v} \tilde{Q}(q_1)\,\tilde{Q}(q_2)\cdots\tilde{Q}(q_k)\;, \tag{S24}$$

i.e., the convolution of k random variables extracted from

$$\tilde{Q}(q) = \frac{(q+1)\mathcal{K}^{(0)}(q+1)}{\langle k\rangle}\;. \tag{S25}$$

This is essentially the same as Eq. (S19), but excluding one edge. It is usually referred as the excess degree distribution [5]. Now, if the number of next-nearest neighbors of the node with degree k is v, then the maximum value of the observed degree is a random variate q obeying the distribution $\mathcal{K}^{(1)}(\leq q|v)$ as defined in Eq. (S20). Thus, the maximum degree of the second-neighbor of a node with degree k obeys the distribution

$$\begin{aligned}\mathcal{K}^{(2)}(\leq q|k) = &\sum_x \mathcal{V}(x|k)\,\mathcal{K}^{(1)}(\leq q|x)\\ &\sum_x \mathcal{V}(x|k)\,\left[Q(\leq q)\right]^x\end{aligned}\;. \tag{S26}$$

For a randomly selected node, we have instead

$$\mathcal{K}^{(2)}(\leq k) = \sum_x \mathcal{K}^{(0)}(x)\,\mathcal{K}^{(2)}(\leq k|x)\;. \tag{S27}$$

Numerical estimates of the distribution $\mathcal{K}^{(2)}(k)$ are displayed in Fig. S5. The expression for the minimum degree between the largest-degree second neighbors of two nodes with degree k_s and k_t is

$$\begin{aligned}&\mathcal{K}^{(2)}_{\min}(\leq k|k_s,k_t) = 1-\left[\mathcal{K}^{(2)}(\leq k|k_s)\right]\left[\mathcal{K}^{(2)}(\leq k|k_t)\right]\\ &= \mathcal{K}^{(2)}(\leq k|k_s)+\mathcal{K}^{(2)}(\leq k|k_t)-\mathcal{K}^{(2)}(\leq k|k_s)\mathcal{K}^{(2)}(\leq k|k_t)\end{aligned}$$

Finally, for the minimum degree between the largest-degree second neighbors of two randomly extracted nodes we have

$$\mathcal{K}^{(2)}_{\min}(\leq k) = 2\mathcal{K}^{(2)}(\leq k)-\left[\mathcal{K}^{(2)}(\leq k)\right]^2\;. \tag{S28}$$

Evaluating the sustainability of the protocols in the configuration model

The distributions $\mathcal{K}^{(x)}_{\min}(\leq k)$ with $x = 0, 1, 2$ determine the actual value of the k characterizing the long-range portion of the communication channel generated by the hxQEP protocol. For $x = 0, 1$, they can be estimated numerically in an efficient way directly from their respective equations; for $x = 2$, estimating the convolution of Eq. (S24) becomes unfeasible already for relatively small networks.

To understand whether the hxQEP protocol is sustainable in the thermodynamic limit in the configuration model, we need to determine how

$$L^{(x)} = \lim_{N\to\infty} \mathcal{K}^{(x)}_{\min}(\leq e^{\ell}) \tag{S29}$$

compares to one. We are looking at the quantity $\mathcal{K}^{(x)}_{\min}(\leq e^{\ell})$ as we know that the redundancy k of a channel should grow at least exponential with ℓ to generate an entangled channel. Then, if the $L^{(x)} = 1$, then no pair of nodes will be able to a have long-range portion of the channel that is able to sustain quantum communication in the limit of infinite graphs, see Eq. (S13). This that the protocol is not sustainable. Otherwise if $L^{(x)} < 1$, the protocol is sustainable in the thermodynamic limit.

Continuous approximation

To compute the above limit, we use a continuous approximation for the various distributions. Accordingly,

$$\mathcal{K}^{(0)}(k) = C^{(0)}\,k^{-\gamma}\;,$$

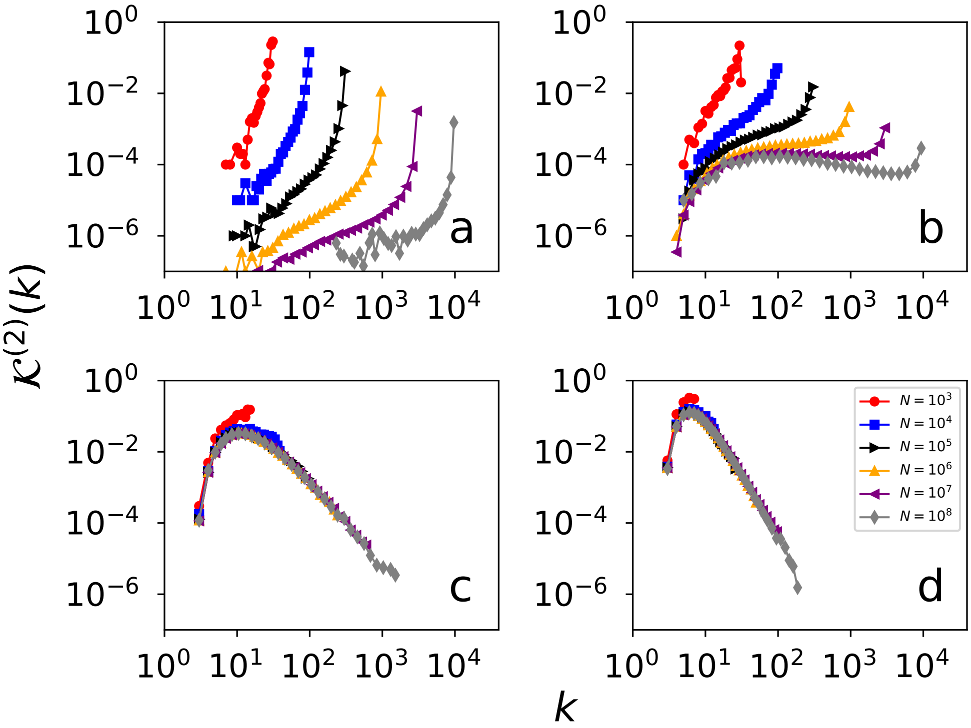

Figure S5. **Statistical properties of the configuration model.** ((a) We plot the distribution of the maximum degree of all nodes at most distance $x = 2$ from a randomly chosen node. This is roughly the same as $\mathcal{K}^{(2)}(k)$. Different colors and symbols are for networks of different sizes N. The degree exponent is $\gamma = 2.1$. Due to the high computational cost, the distribution for $N = 10^8$ is computed over a sample of 10^5 randomly selected nodes. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

where

$$C^{(x)} = \frac{r+1-\gamma}{k_M^{r+1-\gamma} - k_m^{r+1-\gamma}} \ ,$$

thus

$$\mathcal{K}^{(0)}(\leq k) = C^{(0)} \int_{k_m}^{k} dx\, x^{-\gamma} = \frac{k^{1-\gamma} - k_m^{1-\gamma}}{k_M^{1-\gamma} - k_m^{1-\gamma}} \ .$$

In the above equations, k_m is finite, but $k_M = \min\{\sqrt{N}, N^{1/(\gamma-1)}\}$. Also,

$$Q(q) = C^{(1)}\, q^{1-\gamma}$$

thus

$$Q(\leq q) = C^{(1)} \int_{k_m}^{q} x^{1-\gamma} = C^{(1)} \frac{q^{2-\gamma} - k_m^{2-\gamma}}{2-\gamma} = \frac{q^{2-\gamma} - k_m^{2-\gamma}}{k_M^{2-\gamma} - k_m^{2-\gamma}} \ .$$

QEP protocol

We can immediataly write

$$\lim_{N\to\infty} \mathcal{K}^{(0)}(\leq \log N) = \lim_{N\to\infty} \frac{(\log N)^{1-\gamma} - k_m^{1-\gamma}}{k_M^{1-\gamma} - k_m^{1-\gamma}} = 1 \ ,$$

for any $\gamma > 2$, thus from Eq. (S18)

$$\lim_{N\to\infty} \mathcal{K}^{(0)}_{\min}(\leq \log N) = 1 \ .$$

Clearly, this also implies that $\lim_{N\to\infty} \mathcal{K}^{(0)}_{\min}(\leq N^{\alpha}) = 1$. In short, the protocol QEP is not sustainable in the thermodynamic limit.

h1QEP protocol

For the h1QEP protocol, we first notice that

$$\lim_{N\to\infty} Q(\leq \log N) = \frac{(\log N)^{2-\gamma} - k_m^{2-\gamma}}{k_M^{2-\gamma} - k_m^{2-\gamma}} = 1$$

for any $\gamma > 2$. Also, in this case $\lim_{N\to\infty} Q(\leq \log\log N) = 1$. Using Eq. (S23), we have that

$$\lim_{N\to\infty} \mathcal{K}^{(1)}_{\min}(\leq \log N) = 1 \ ,$$

thus the protocol h1QEP is not sustainable in the thermodynamic limit.

h2QEP protocol

For the h2QEP protocol, we approximate

$$\mathcal{V}(v|k) \simeq \delta\left(v, k\frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}\right)$$

where the Kronecker delta function is such that $\delta(x, y) = 1$ if $x = y$ and $\delta(x, y) = 0$ otherwise. The term $k\frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}$ is the average value of the convolution of Eq. (S24), see Eq. (S34). We stress that this is a rather loose approximation as the standard deviation of the distribution grows even faster than its average as the network size grows, see Eq. (S36). We can then write

$$\mathcal{K}^{(2)}(\leq q|k) = \left\{[Q(\leq q)]^{(\langle k^2\rangle - \langle k\rangle)/\langle k\rangle}\right\}^k$$

thus

$$\mathcal{K}^{(2)}_{\min}(\leq k) = 2G_0\left[\mathcal{F}(\leq q)\right] - \left\{G_0\left[\mathcal{F}(\leq q)\right]\right\}^2$$

where

$$\mathcal{F}(\leq q) = [Q(\leq q)]^{(\langle k^2\rangle - \langle k\rangle)/\langle k\rangle} \ .$$

Now, we consider

$$\lim_{N\to\infty} \mathcal{F}(\leq \log N) = \lim_{N\to\infty} [Q(\leq \log N)]^{\langle k^2\rangle/\langle k\rangle - 1} \ .$$

For $\gamma > 3$, $\langle k^2\rangle$ is finite and the above limit goes to one. For $2 < \gamma \leq 3$, we have instead

$$\lim_{N\to\infty} \mathcal{F}(\leq \log N) = 0$$

as $Q(\leq \log N) = 1 - Q(> \log N) \sim 1 - (\log N)^{2-\gamma}$ and $\langle k^2\rangle \sim N^{\frac{3-\gamma}{2}}$. In summary,

$$\lim_{N\to\infty} \mathcal{K}^{(2)}(\leq \log N) = \begin{cases} 0 & \text{if } 2 < \gamma \leq 3 \\ 1 & \text{if } \gamma > 3 \end{cases} \ ,$$

showing that the h2QEP protocol is sustainable in the thermodynamic limit for scale-free graphs with degree exponent $2 < \gamma \leq 3$. We notice that the very same argument can also be used to show that

$$\lim_{N\to\infty} \mathcal{K}^{(2)}(\leq N^{\alpha}) = \begin{cases} 0 & \text{if } 2 < \gamma \leq 3 \\ 1 & \text{if } \gamma > 3 \end{cases} \ ,$$

for $\alpha < \frac{3-\gamma}{2}$, i.e., N^{α} grows to infinity slower than $\langle k^2\rangle$.

Generating function formalism

We rely on the fact that if a generic distribution $\mathcal{R}(k)$ has generating function $R(z) = \sum_k z^k \mathcal{R}(k)$, then we trivially have the normalization condition

$$R(z)|_{z=1} = R(1) = 1 \,, \tag{S30}$$

and the first three derivatives of the generating function are related to the first three moments of the distribution respectively as

$$z\,R'(z)\Big|_{z=1} = R'(1) = \langle r\rangle \,, \tag{S31}$$

$$z^2\,R''(x)\Big|_{z=1} = R''(1) = \langle r^2\rangle - \langle r\rangle \,, \tag{S32}$$

and

$$z^3\,R'''(z)\Big|_{z=1} = R'''(1) = \langle r^3\rangle - \langle r^2\rangle + 2\langle r\rangle \,. \tag{S33}$$

We use the above identities to evaluate moments of the other probability distributions in term of the moments of the degree distribution $\mathcal{K}^{(0)}(k)$, see Eqs. (S30), (S31), (S32), and (S33). To this end, we define the generating function of the degree distribution

$$G_0(z) = \sum_k z^k\,\mathcal{K}^{(0)}(k) \,.$$

We note that the distribution of Eq. (S25) is generated by

$$G_1(z) = \sum_k z^k \tilde{Q}(k) = \sum_k z^{k-1}\,\frac{k}{\langle k\rangle}\mathcal{K}^{(0)}(k) = \frac{G_0'(z)}{G_0'(1)} \,.$$

If we take the derivative with respect to z, we have

$$G_1'(z) = \frac{G_0''(z)}{G_0'(1)} \,,$$

and if we take another derivative, we get

$$G_1''(z) = \frac{G_0'''(z)}{G_0'(1)} \,.$$

We note that the generating function of the distribution of Eq. (S24) is

$$\sum_v z^v\,\mathcal{V}(v|k) = [G_1(z)]^k \,.$$

The first moment of such a distribution is

$$\langle v\rangle = \sum_v v\,\mathcal{V}(v|k) = z\,\frac{d}{dz}\,[G_1(z)]^k\Big|_{z=1} \,,$$

and the second moment is

$$\langle v^2\rangle = \sum_v v^2\,\mathcal{V}(v|k) = z^2\,\frac{d^2}{dz^2}\,[G_1(z)]^k\Big|_{z=1} + \langle v\rangle \,.$$

We note that

$$z\,\frac{d}{dz}\,[G_1(z)]^k\Big|_{z=1} = k\,G_1'(1)\,[G_1(1)]^{k-1} = k\,G_1'(1) \,,$$

which leads to

$$\langle v\rangle = k\,\frac{G_0''(1)}{G_0'(1)} = k\,\frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle} \,. \tag{S34}$$

Also,

$$\begin{array}{l} z^2 \frac{d^2}{dz^2} \left. [G_1(z)]^k \right|_{z=1} = \\ k\,[G_1(1)]^{k-2} \left\{ G_1''(1) G_1(1) + (k-1) \left[G_1'(1) \right]^2 \right\} = \\ k \left\{ G_1''(1) + (k-1) \left[G_1'(1) \right]^2 \right\} = \\ k \left\{ G_0'''(1)/G_0'(1) + (k-1) \left[G_0''(1)/G_0'(1) \right]^2 \right\} \end{array} .$$

We have that

$$\langle v^2 \rangle - \langle v \rangle = \frac{k}{\langle k \rangle} \left\{ \langle k^3 \rangle - \langle k^2 \rangle + 2\langle k \rangle + (k-1) \frac{\left[\langle k^2 \rangle - \langle k \rangle \right]^2}{\langle k \rangle} \right\} . \tag{S35}$$

We note that the coefficient of variation of the distribution $\mathcal{V}(v|k)$ is

$$\frac{\sqrt{\langle v^2 \rangle - \langle v \rangle}}{\langle v \rangle} \simeq \frac{1}{\sqrt{k}} \frac{\sqrt{\langle k^3 \rangle}}{\langle k^2 \rangle} , \tag{S36}$$

if one considers the highest-order moments only in the expressions of Eqs. (S34) and (S35).

Quantum communication protocols on the configuration model

In Figs. S6, S7, S8 and S9, we display the diagram of the various communication protocols applied to instances of the CM. Results for QEP and CEP are based on averages over $P = 10000$ pairs of randomly selected pairs of nodes. However, for $N = 10^8$, we consider $P = 1000$ pairs. For the h1QEP, we use at least $P = 1000$, except for the case $N = 10^8$ where results rely on $P = 100$. Simulations for $N = 10^8$ and $\gamma = 2.1$ are not computationally feasible. Finally, for the h2QEP, we have at least $P = 1000$ for $N \leq 10^5$, $P = 100$ for $N = 10^6, 10^7$. Simulations for $N = 10^8$ are feasible only for $\gamma = 4.5$ where we use $P = 20$.

An estimate of the time T required to run the various protocols on networks with size N is provided in Fig. S10. Here T is estimated in the nearly worst-case scenario $\lambda = 0.99$, so that all paths between nodes s and t are exploited by the protocols. Clearly, for better entangled networks, the time required to run the protocols is shorter. We see a clear scaling

$$T \sim N^\eta \tag{S37}$$

where the best estimate of η depends on the protocol when $2 < \gamma \leq 3$, but becomes protocol independent for $\gamma > 3$.

Quantum communication protocols on $\mathbb{S}^1$ model

We consider also instances of the $\mathbb{S}^1$ model [19]. In that, we first assign to each node i a degree k_i by extracting a random variate from Eq. (S17). Then, we also assign to each node i the angular coordinate θ_i uniformly at random over the interval $[0, 2\pi)$. Then, we connect each pair of nodes i and j with probability

$$p_{i,j} = \frac{1}{1 + \left(\frac{R\,\Delta\theta_{i,j}}{\mu k_i k_j} \right)^{-\beta}} , \tag{S38}$$

where $R = \frac{2\pi}{N}$ and $\mu = \frac{\beta-1}{2\langle k \rangle}$. In particular, $\langle k \rangle = \frac{1}{N} \sum_i k_i$, and $\Delta\theta_{i,j} = \min\{|\theta_i - \theta_j|, 2\pi - |\theta_i - \theta_j|\}$. We consider only the case $\beta = 1.5$, which creates graphs with average clustering coefficient roughly equal to $c = 0.1$ (the specific value of c depends on the network size N and the degree exponent γ).

Our numerical results are based on averages over 5 instances of the $\mathbb{S}^1$ for the various combinations of the parameters N and γ, except for $N = 10^7$ where we consider only 1 instance of the model. Unfortunately, the quadratic time complexity of the generative algorithm did not allow us to consider larger network sizes. When estimating the performance of the various protocols, we consider at least $P = 1000$ pairs of randomly chosen nodes.

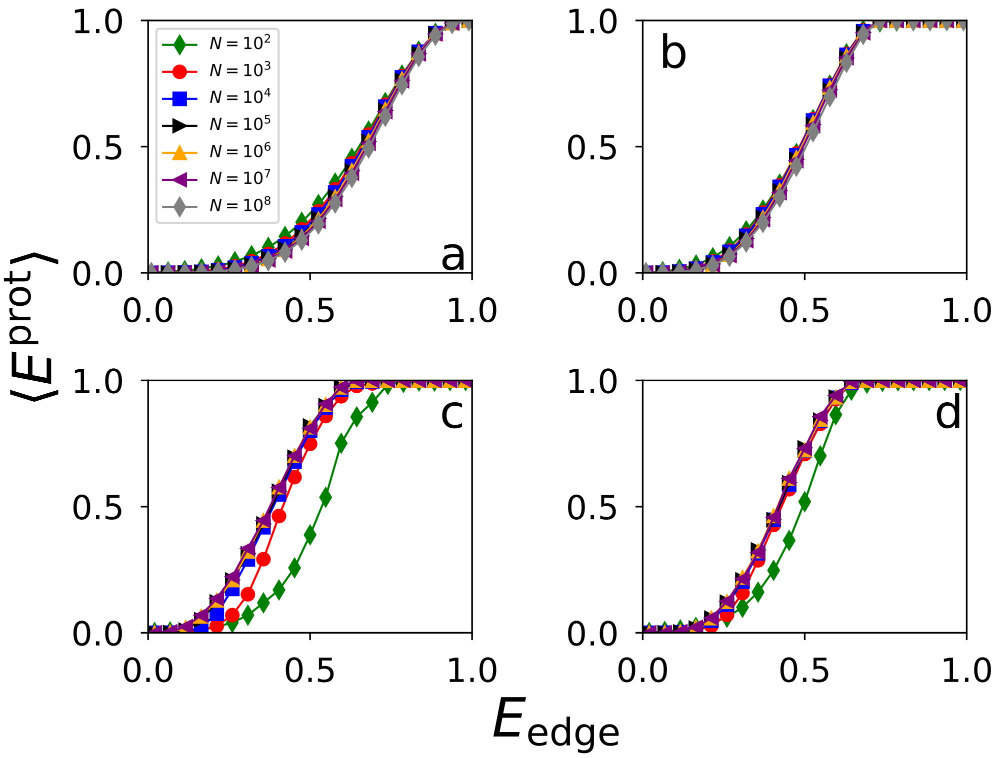

Figure S6. **Quantum communication on synthetic networks.** (a) We plot the average entanglement $\langle E^{\mathrm{CEP}} \rangle$ as a function of the entanglement of the individual edges E_{edge} for network generated according to the configuration model with degree exponent $\gamma = 2.1$. Different symbols and colors correspond to different network sizes. (b-d) Same as in (a), but for QEP, h1QEP and h2QEP, respectively.

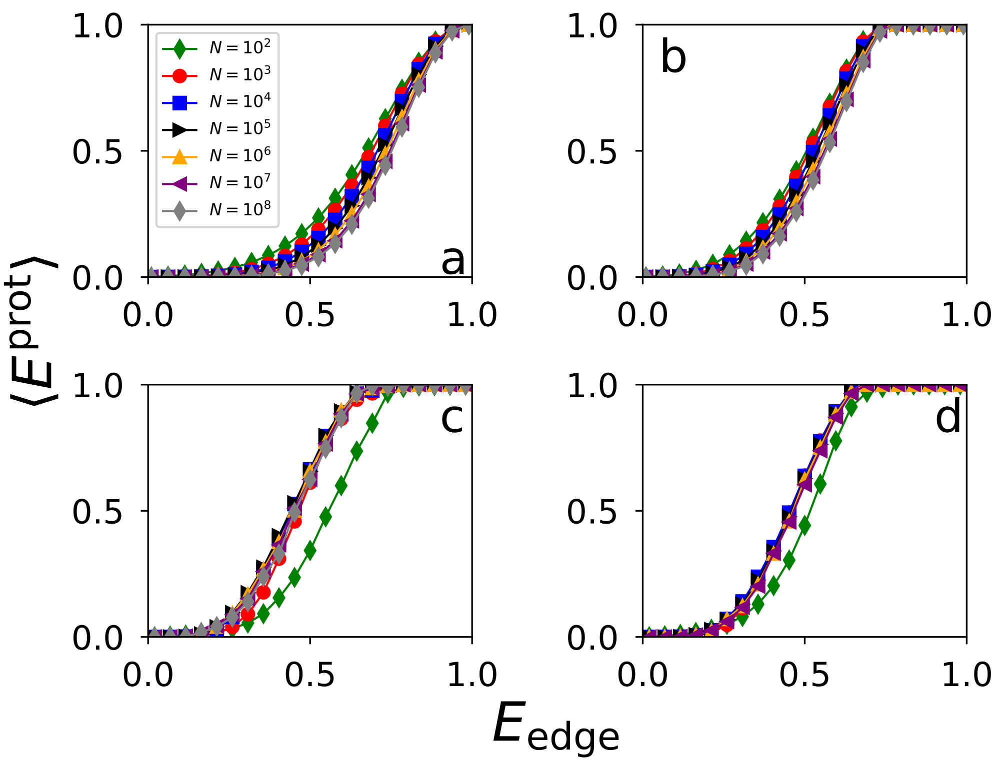

Figure S7. **Quantum communication on synthetic networks.** Same as in Fig. S6 but for $\gamma = 2.5$.

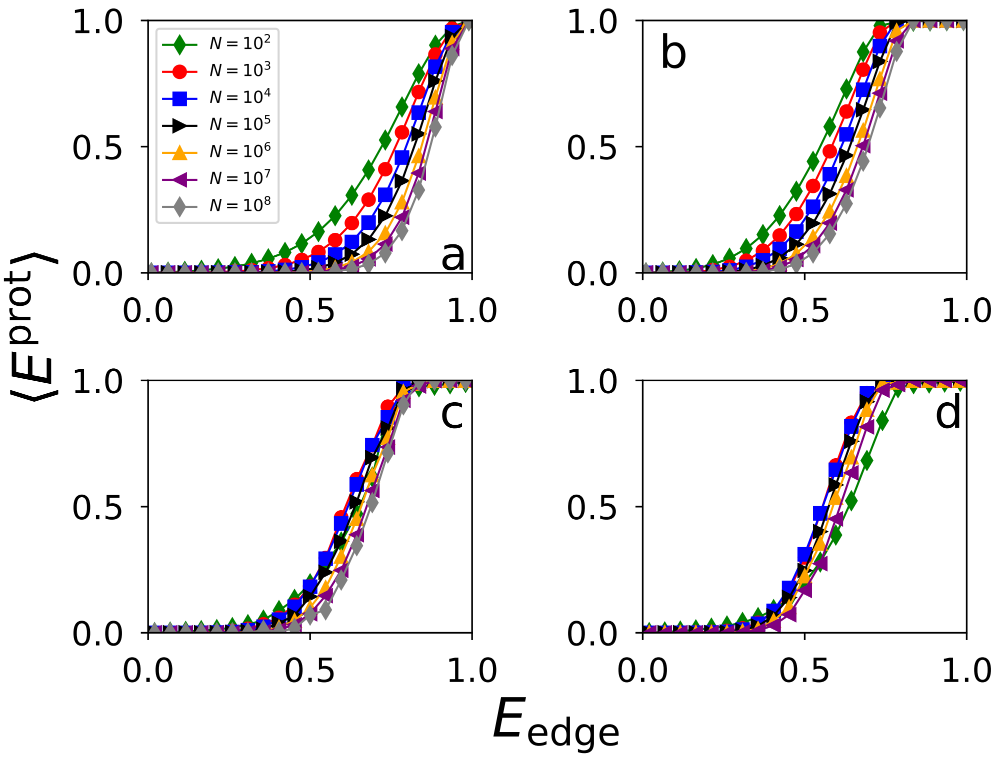

Figure S8. **Quantum communication on synthetic networks.** Same as in Fig. S6 but for $\gamma = 3.5$.

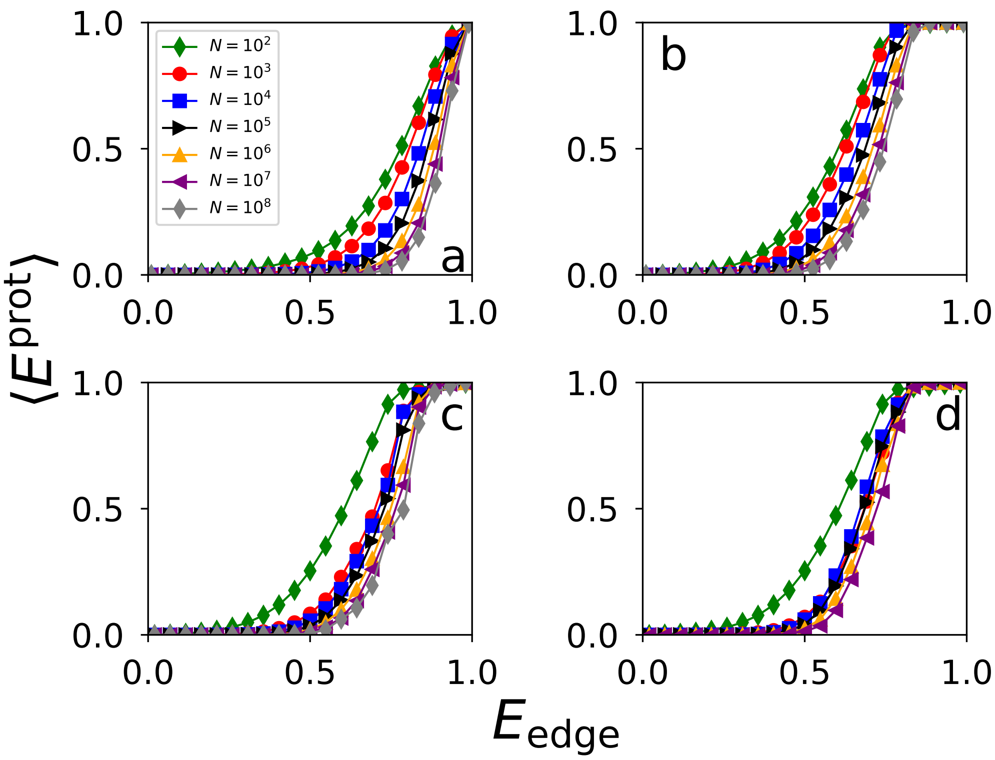

Figure S9. **Quantum communication on synthetic networks.** Same as in Fig. S6 but for $\gamma = 4.5$.

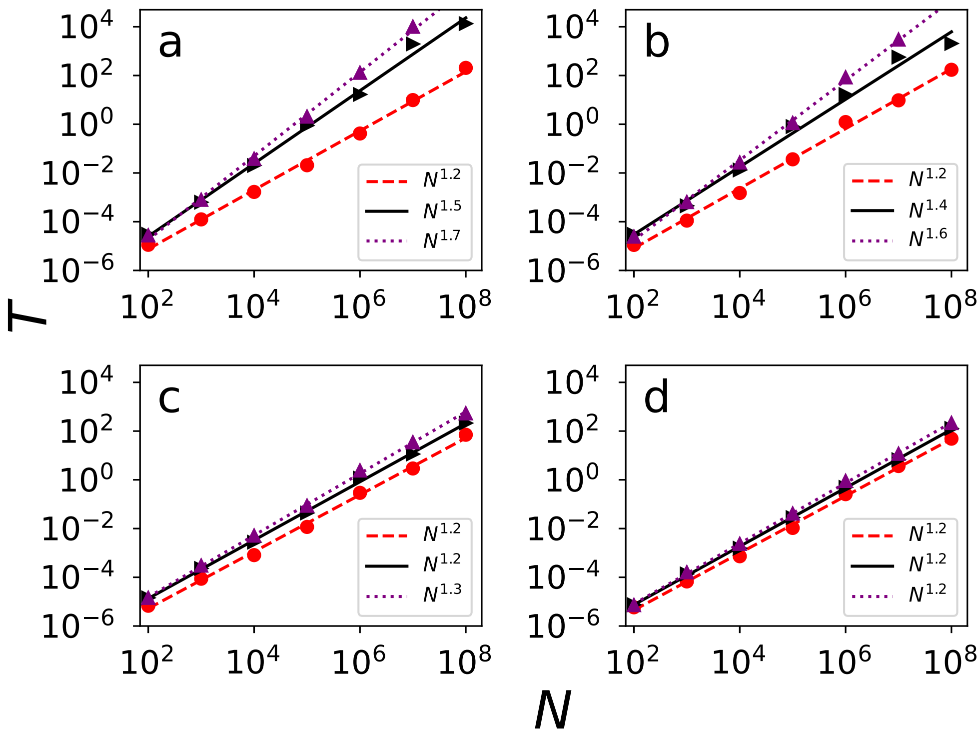

Figure S10. **Computational complexity of the communication protocols.** (a) Time T required to find all possible paths between two randomly selected nodes s and t for the QEP/CEP (red circles), h1QEP (black triangles) and the h2QEP (purple triangles) protocols. Computations were performed with a `INTEL(R) XEON(R) GOLD 6548N` cpu. Time is measured in seconds, and the results are averaged over at least 10 randomly selected pairs. Results are valid for the configuration model with $\gamma = 2.1$. T is plotted as a function of the network size N. The displayed lines represent best power-law fits $T \sim N^{\eta}$ of the data points. We find $\eta = 1.2$ for QEP, $\eta = 1.5$ for h1QEP, and $\eta = 1.7$ for h2QQEP. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively. The best estimates of the power-law exponents are reported in the legends of the various plots.

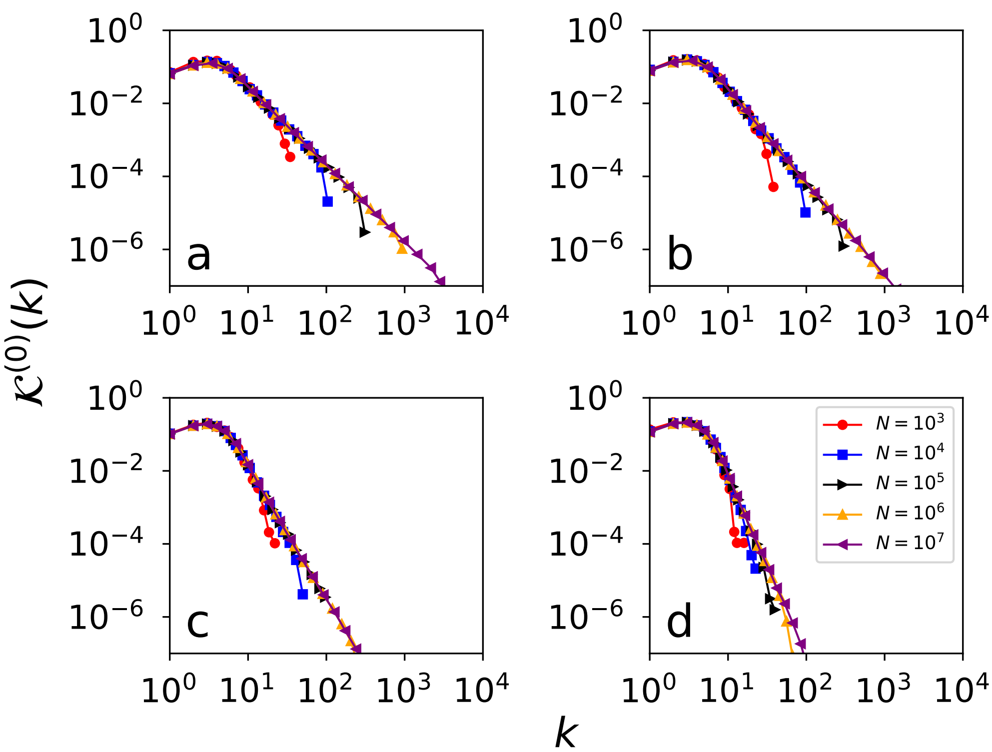

Figure S11. **Statistical properties of the** $\mathbb{S}^1$ **model.** (a) We plot the degree distribution $\mathcal{K}^{(0)}(k)$ for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

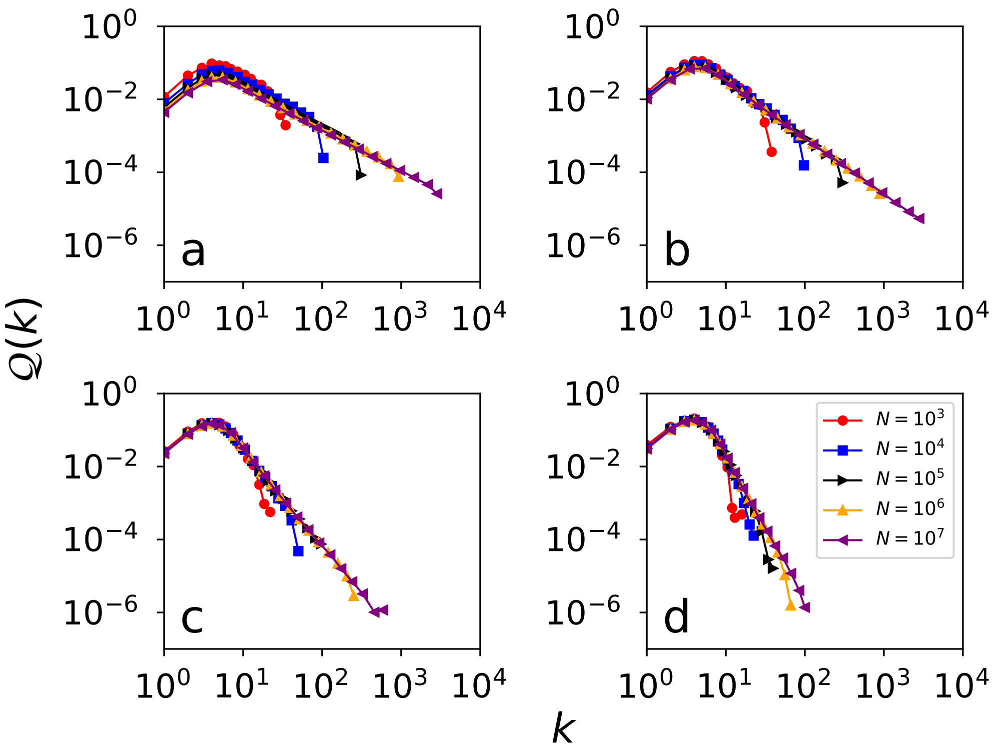

Figure S12. **Statistical properties of the** $\mathbb{S}^1$ **model.** (a) We plot the distribution $\mathcal{Q}(k)$ for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

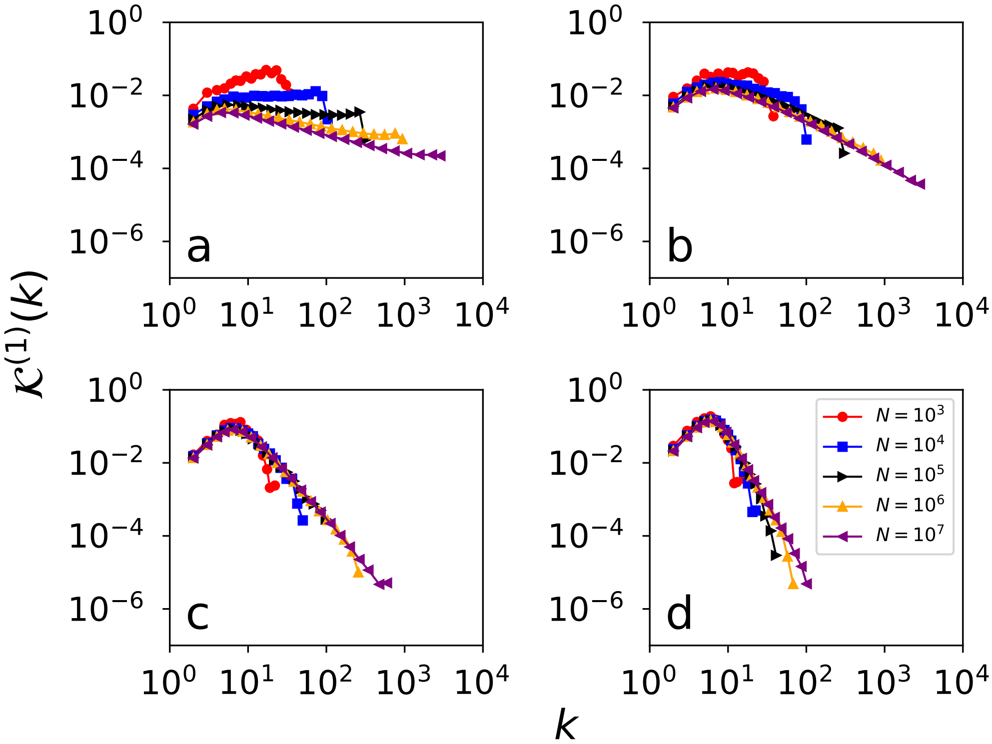

Figure S13. **Statistical properties of the $\mathbb{S}^1$ model.** (a) We plot the distribution of the maximum degree of all nodes at most distance $x = 1$ from a randomly chosen node. This is roughly the same as $\mathcal{K}^{(1)}(k)$. Different colors and symbols are for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

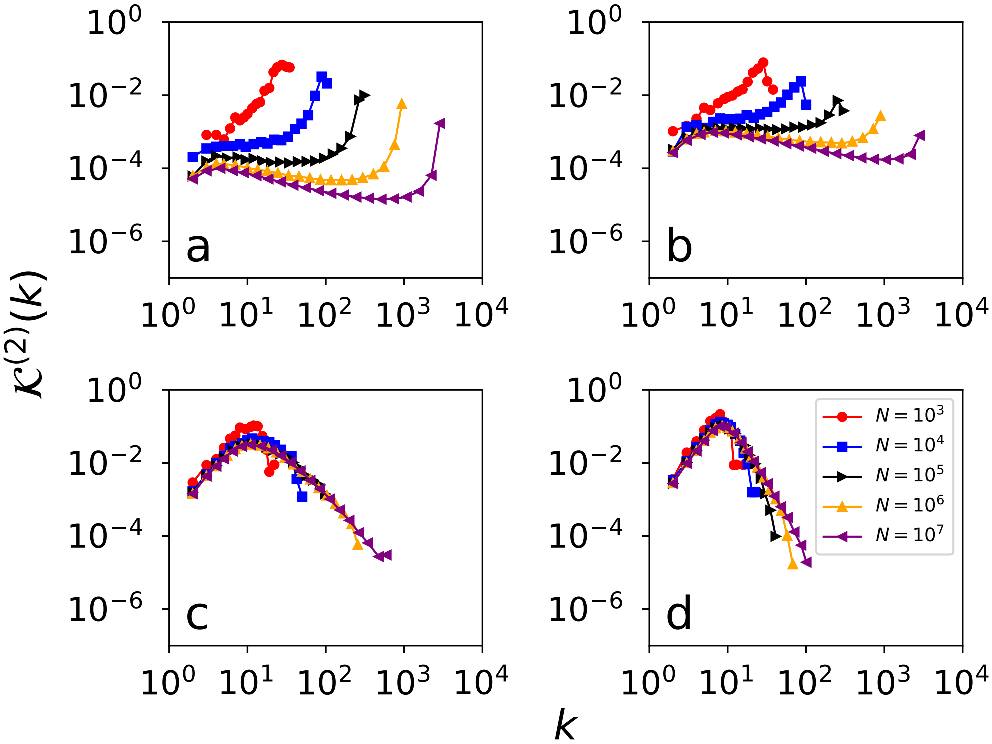

Figure S14. **Statistical properties of the $\mathbb{S}^1$ model.** (a) We plot the distribution of the maximum degree of all nodes at most distance $x = 2$ from a randomly chosen node. This is roughly the same as $\mathcal{K}^{(2)}(k)$. Different colors and symbols are for networks of different sizes N. The degree exponent is $\gamma = 2.1$. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

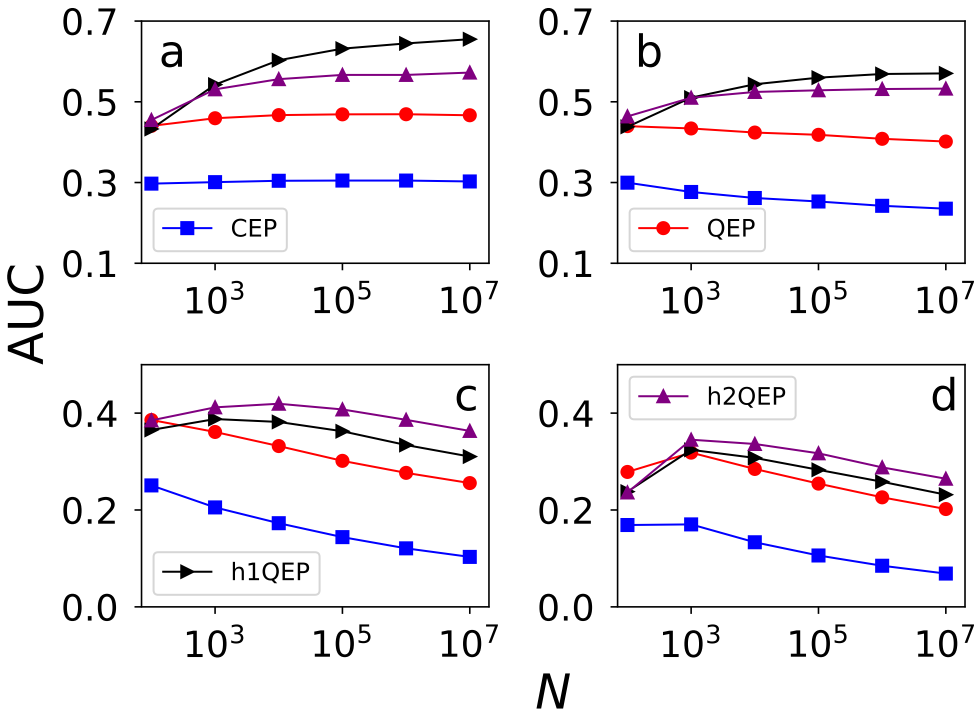

Figure S15. **Protocols of quantum communication on $\mathbb{S}^1$ model.** (a) We measure the Area Under the Curve (AUC) for the various communication protocols on graphs generated using to the $\mathbb{S}^1$ model. Here, the degree distribution is a power-law exponent $\gamma = 2.1$. The AUC metric is plotted as a function of the network size N. Results are averaged over multiple instances of the network model, and many randomly selected pairs of nodes s and t, see SM for details. (b-d) Same as in (a), but for $\gamma = 2.5, 3.5$ and 4.5, respectively.

REAL NETWORKS

In Tables S1 and S2, we provide a complete list of the real networks analyzed. Individual diagrams for the various networks are shown in Figs. S16, S17, S18 and S19. Results are obtained using $P = 10000$ for QEP and CEP, and $P = 1000$ for h1QEP and h2QEP.

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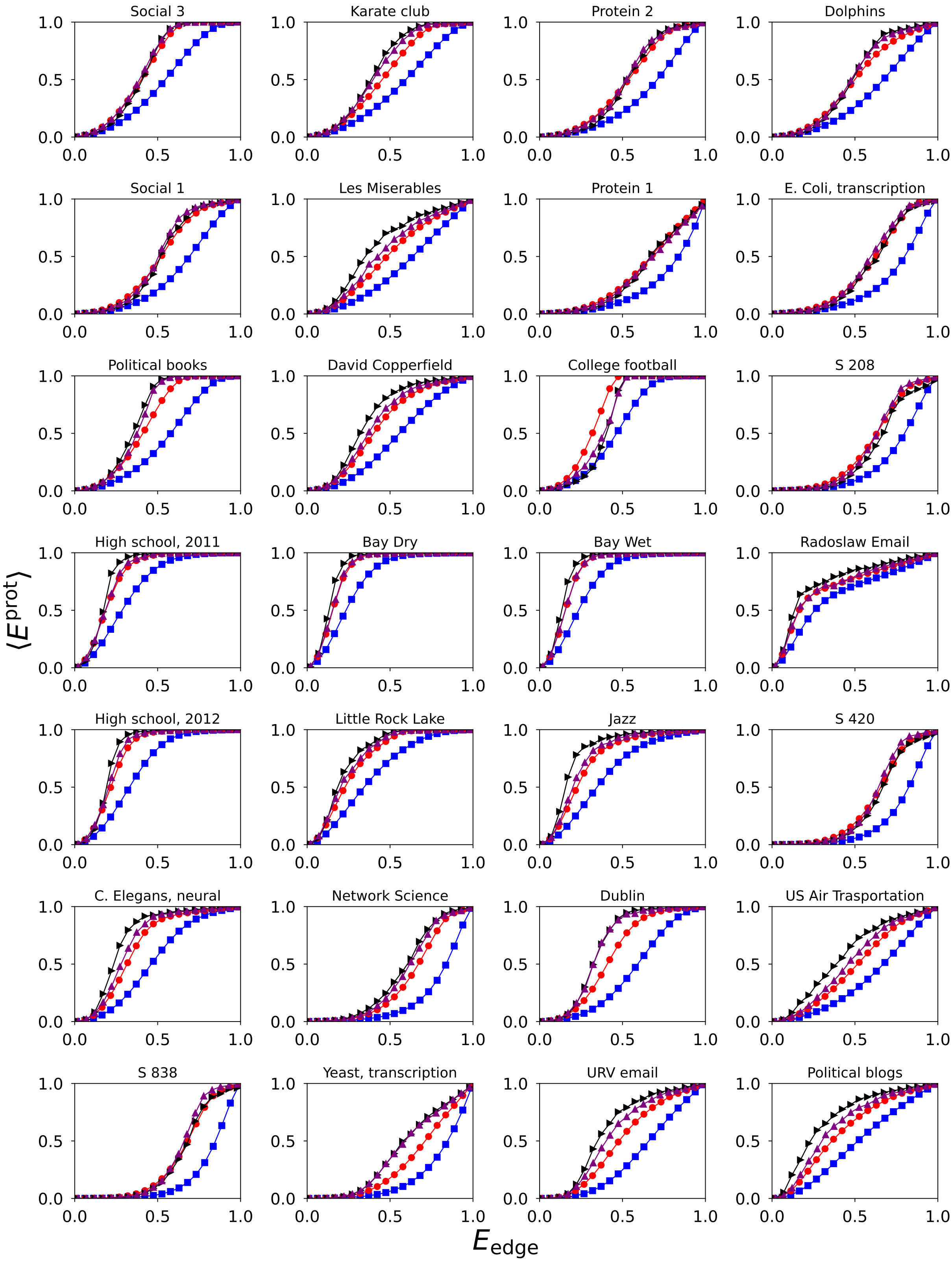

Figure S16. **Quantum communication on real networks.** We plot the average entanglement $\langle E^{\text{prot}} \rangle$ as a function of the entanglement of the individual edges E_{edge} for the real networks listed in Tables S1 and S2. We use blue squares for the CEP protocol, red circles for QEP, black triangles for h1QEP, and purple triangles for h2QEP. Each data point is obtained by averaging over $P = 10000$ random pairs of nodes for CEP and QEP, and $P = 1000$ for h1QEP and h2QEP.

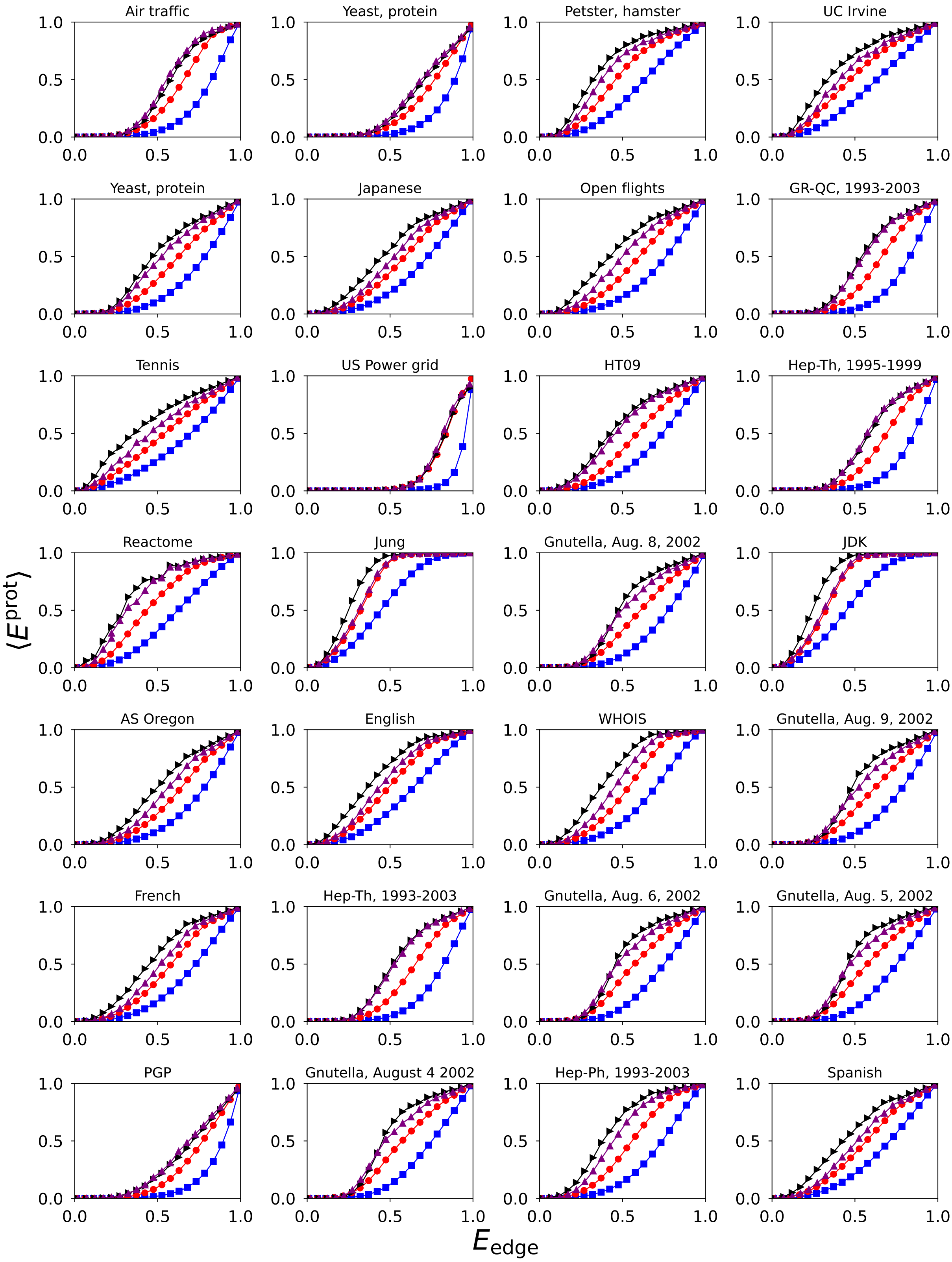

Figure S17. **Quantum communication on real networks.** Continuation of Fig. S16.

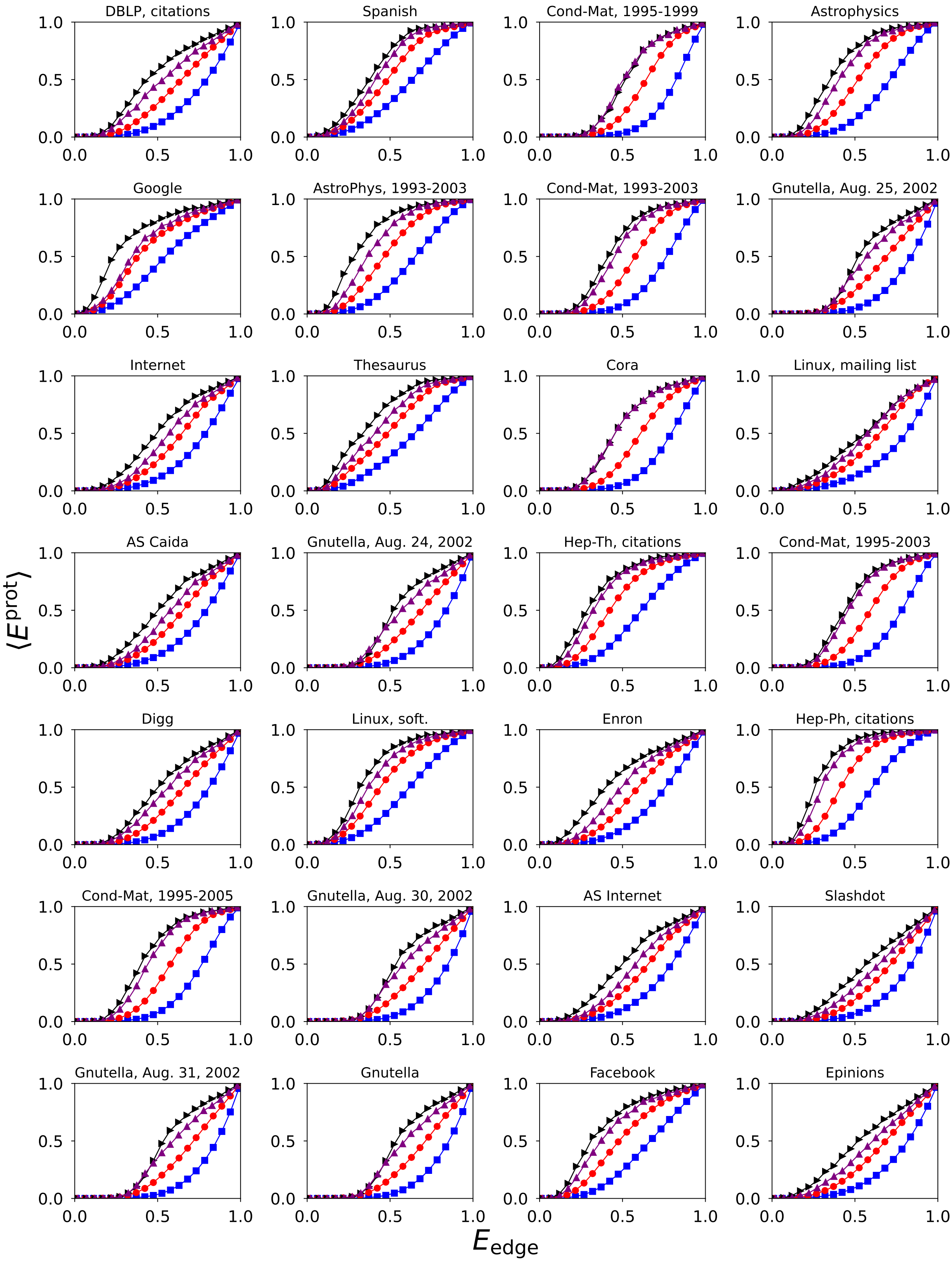

Figure S18. **Quantum communication on real networks.** Continuation of Fig. S17.

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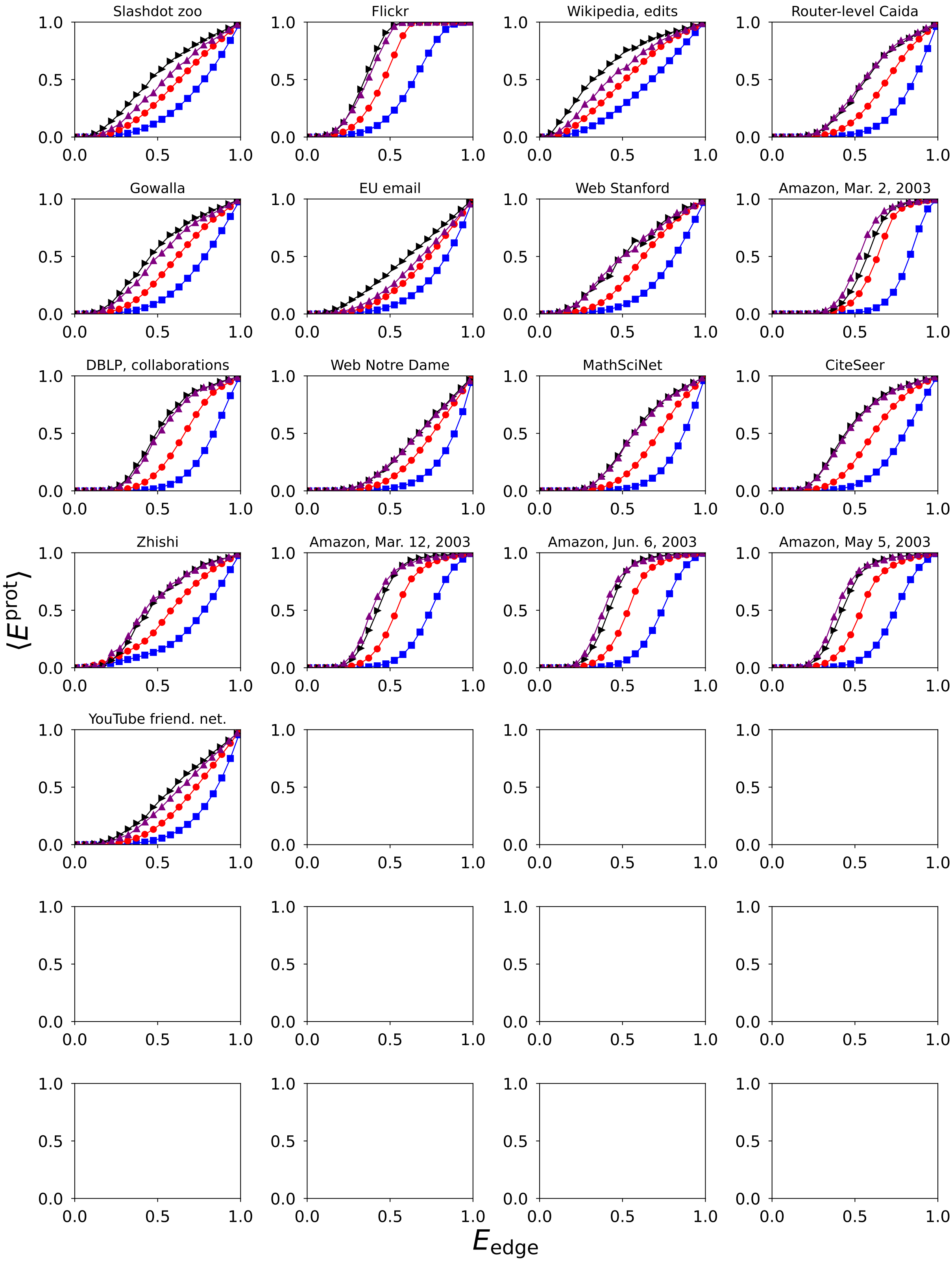

Figure S19. **Quantum communication on real networks.** Continuation of Fig. S18.

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Network	N	M	$\langle k^2\rangle/\langle k\rangle^2$	AUC$^{\mathrm{CEP}}$	AUC$^{\mathrm{QEP}}$	AUC$^{\mathrm{h1QEP}}$	AUC$^{\mathrm{h2QEP}}$	Refs.	URL
Social 3	32	80	1.19	0.47	0.61	0.61	0.62	[20]	URL
Karate club	34	78	1.69	0.43	0.55	0.61	0.60	[21]	URL
Protein 2	53	123	1.16	0.33	0.48	0.47	0.49	[20]	URL
Dolphins	62	159	1.33	0.36	0.49	0.51	0.52	[22]	URL
Social 1	67	142	1.24	0.34	0.48	0.48	0.50	[20]	URL
Les Miserables	77	254	1.83	0.39	0.50	0.59	0.54	[23]	URL
Protein 1	95	213	1.10	0.22	0.35	0.33	0.33	[20]	URL
E. Coli, transcription	97	212	1.24	0.25	0.40	0.39	0.42	[24]	URL
Political books	105	441	1.42	0.44	0.60	0.66	0.64	[25]	URL
David Copperfield	112	425	1.81	0.45	0.57	0.65	0.60	[26]	URL
College football	115	613	1.01	0.55	0.70	0.62	0.64	[27]	URL
S 208	122	189	1.22	0.24	0.38	0.34	0.38	[20]	URL
High school, 2011	126	1709	1.21	0.70	0.80	0.83	0.81	[28]	URL
Bay Dry	128	2106	1.23	0.75	0.84	0.86	0.84	[29, 30]	URL
Bay Wet	128	2075	1.24	0.75	0.84	0.86	0.84	[30]	URL
Radoslaw Email	167	3250	1.66	0.64	0.71	0.76	0.72	[30, 31]	URL
High school, 2012	180	2220	1.20	0.66	0.78	0.81	0.79	[28]	URL
Little Rock Lake	183	2434	1.61	0.63	0.74	0.78	0.77	[30, 32]	URL
Jazz	198	2742	1.40	0.62	0.74	0.81	0.76	[33]	URL
S 420	252	399	1.23	0.21	0.36	0.34	0.37	[20]	URL
C. Elegans, neural	297	2148	1.80	0.53	0.66	0.74	0.70	[34]	URL
Network Science	379	914	1.66	0.20	0.35	0.41	0.39	[26]	URL
Dublin	410	2765	1.39	0.42	0.58	0.66	0.66	[30, 35]	URL
US Air Trasportation	500	2980	4.51	0.36	0.48	0.59	0.53	[36]	URL
S 838	512	819	1.26	0.18	0.33	0.33	0.35	[20]	URL
Yeast, transcription	662	1062	4.21	0.20	0.31	0.39	0.39	[37]	URL
URV email	1133	5451	1.94	0.35	0.50	0.60	0.56	[38]	URL
Political blogs	1222	16714	2.97	0.47	0.59	0.70	0.64	[25]	URL
Air traffic	1226	2408	1.87	0.21	0.36	0.41	0.43	[30]	URL
Yeast, protein	1458	1948	2.67	0.15	0.26	0.29	0.31	[39]	URL
Petster, hamster	1788	12476	3.26	0.38	0.51	0.63	0.58	[30]	URL
UC Irvine	1893	13835	3.81	0.38	0.50	0.61	0.55	[30, 40]	URL
Yeast, protein	2224	6609	2.80	0.25	0.38	0.49	0.45	[41]	URL
Japanese	2698	7995	18.33	0.31	0.42	0.52	0.47	[20]	URL
Open flights	2905	15645	5.18	0.28	0.41	0.55	0.48	[30, 42]	URL
GR-QC, 1993-2003	4158	13422	2.79	0.20	0.34	0.44	0.43	[43]	URL
Tennis	4338	81865	4.21	0.34	0.46	0.59	0.52	[44]	URL
US Power grid	4941	6594	1.45	0.07	0.18	0.18	0.19	[34]	URL
HT09	5352	18481	28.96	0.29	0.41	0.54	0.52	[35]	URL
Hep-Th, 1995-1999	5835	13815	1.93	0.17	0.32	0.40	0.41	[45]	URL
Reactome	5973	145778	2.94	0.39	0.54	0.66	0.63	[30, 46]	URL
Jung	6120	50290	60.35	0.55	0.68	0.74	0.69	[30, 47]	URL
Gnutella, Aug. 8, 2002	6299	20776	2.68	0.25	0.39	0.48	0.46	[43, 48]	URL
JDK	6434	53658	58.92	0.55	0.68	0.75	0.69	[30]	URL
AS Oregon*	6474	12572	42.43	0.26	0.37	0.48	0.43	[49]	URL
English	7377	44205	26.76	0.38	0.50	0.61	0.54	[20]	URL
WHOIS*	7476	56943	8.91	0.33	0.47	0.62	0.54	[50, 51]	URL
Gnutella, Aug. 9, 2002	8104	26008	2.62	0.24	0.38	0.48	0.45	[43, 48]	URL
French	8308	23832	38.00	0.30	0.42	0.53	0.47	[20]	URL
Hep-Th, 1993-2003	8638	24806	2.26	0.21	0.35	0.46	0.45	[43]	URL
nutella, Aug. 6, 2002	8717	31525	1.99	0.27	0.41	0.50	0.49	[43, 48]	URL
Gnutella, Aug. 5, 2002	8842	31837	2.05	0.27	0.41	0.50	0.49	[43, 48]	URL
PGP	10680	24316	4.15	0.14	0.25	0.31	0.32	[52]	URL
Gnutella, August 4 2002	10876	39994	1.90	0.27	0.41	0.50	0.48	[43, 48]	URL
Hep-Ph, 1993-2003	11204	117619	6.23	0.29	0.44	0.59	0.53	[43]	URL
Spanish	11558	43050	61.43	0.32	0.43	0.54	0.47	[20]	URL
DBLP, citations	12495	49563	5.52	0.24	0.37	0.51	0.45	[30, 53]	URL
Spanish	12643	55019	92.80	0.38	0.51	0.61	0.58	[30]	URL
Cond-Mat, 1995-1999	13861	44619	2.10	0.20	0.36	0.46	0.47	[45]	URL
Astrophysics	14845	119652	2.82	0.31	0.47	0.62	0.57	[45]	URL
Google	15763	148585	47.83	0.44	0.56	0.69	0.60	[54]	URL
AstroPhys, 1993-2003	17903	196972	2.99	0.36	0.52	0.67	0.61	[43]	URL

Table S1. **Real networks.** From left to right, for each network we report: name, number of nodes, number of edges, ratio of the second moment of the degree distribution and the square of the first moment, AUC metrics for the protocols CEP, QEP, h1QEP and h2QEP, paper(s) where the dataset was first studied, and URL of where data were retrieved. We consider only the largest connected component of each network. Snapshot of the Internet are highlighted by an asterisk close to the network name.

Network	N	M	$\langle k^2\rangle/\langle k\rangle^2$	AUC$^{\mathrm{CEP}}$	AUC$^{\mathrm{QEP}}$	AUC$^{\mathrm{h1QEP}}$	AUC$^{\mathrm{h2QEP}}$	Refs.	URL
Cond-Mat, 1993-2003	21363	91286	2.63	0.26	0.42	0.56	0.53	[43]	URL
Gnutella, Aug. 25, 2002	22663	54693	2.23	0.20	0.32	0.43	0.40	[43, 48]	URL
Internet	22963	48436	61.98	0.25	0.37	0.49	0.42	-	URL
Thesaurus	23132	297094	4.02	0.38	0.52	0.64	0.58	[30, 55]	URL
Cora	23166	89157	3.08	0.23	0.39	0.51	0.51	[30, 56]	URL
Linux, mailing list	24567	158164	26.48	0.25	0.37	0.45	0.43	[30]	URL
AS Caida*	26475	53381	69.50	0.25	0.36	0.48	0.41	[49]	URL
Gnutella, Aug. 24, 2002	26498	65359	2.44	0.20	0.32	0.44	0.41	[43, 48]	URL
Hep-Th, citations	27400	352021	4.14	0.39	0.55	0.67	0.64	[30, 43]	URL
Cond-Mat, 1995-2003	27519	116181	2.64	0.24	0.40	0.54	0.52	[45]	URL
Digg	29652	84781	4.91	0.22	0.35	0.47	0.42	[30, 57]	URL
Linux, soft.	30817	213208	61.62	0.39	0.53	0.64	0.60	[30]	URL
Enron	33696	180811	13.27	0.27	0.40	0.55	0.48	[58]	URL
Hep-Ph, citations	34401	420784	2.60	0.39	0.56	0.71	0.67	[30, 43]	URL
Cond-Mat, 1995-2005	36458	171735	2.96	0.25	0.42	0.57	0.53	[45]	URL
Gnutella, Aug. 30, 2002	36646	88303	2.38	0.19	0.31	0.43	0.39	[43, 48]	URL
AS Internet*	40164	85123	79.53	0.24	0.36	0.48	0.41	[50, 59]	URL
Slashdot	51083	116573	17.87	0.20	0.31	0.42	0.37	[30, 60]	URL
Gnutella, Aug. 31, 2002	62561	147878	2.45	0.18	0.30	0.42	0.39	[43, 48]	URL
Gnutella	62561	147878	2.45	0.18	0.30	0.42	0.39	[50, 61]	URL
Facebook	63392	816886	3.42	0.34	0.49	0.63	0.58	[62]	URL
Epinions	75877	405739	17.19	0.23	0.33	0.46	0.40	[30, 63]	URL
Slashdot zoo	79116	467731	12.37	0.26	0.38	0.51	0.44	[30, 64]	URL
Flickr	105722	2316668	7.97	0.37	0.55	0.66	0.64	[30, 65]	URL
Wikipedia, edits	113123	2025910	19.25	0.35	0.48	0.62	0.54	[30, 66]	URL
Router-level Caida*	190914	607610	5.96	0.17	0.31	0.42	0.43	[50, 59, 67]	URL
Gowalla	196591	950327	31.71	0.25	0.38	0.51	0.47	[30, 68]	URL
EU email	224832	339925	187.73	0.20	0.30	0.41	0.33	[30, 43]	URL
Web Stanford	255265	1941926	133.47	0.22	0.37	0.47	0.47	[58]	URL
Amazon, Mar. 2, 2003	262111	899792	1.62	0.18	0.37	0.43	0.47	[69]	URL
DBLP, collaborations	317080	1049866	3.28	0.19	0.34	0.48	0.46	[30, 53]	URL
Web Notre Dame	325729	1090108	41.93	0.15	0.26	0.34	0.33	[70]	URL
MathSciNet	332689	820644	3.33	0.16	0.29	0.41	0.41	[71]	URL
CiteSeer	365154	1721981	5.14	0.22	0.38	0.52	0.51	[30, 72]	URL
Zhishi	372840	2318025	2244.54	0.27	0.42	0.51	0.53	[30, 73]	URL
Amazon, Mar. 12, 2003	400727	2349869	2.59	0.26	0.45	0.56	0.58	[69]	URL
Amazon, Jun. 6, 2003	403364	2443311	2.52	0.27	0.46	0.57	0.59	[69]	URL
Amazon, May 5, 2003	410236	2439437	2.60	0.26	0.45	0.56	0.59	[69]	URL
YouTube friend. net.	1134890	2987624	93.93	0.18	0.29	0.40	0.36	[30, 74]	URL

Table S2. **Real networks.** Continuation of Table S1

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